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CHAPTER 5

CONFIGURATION OF NETWORK TOPOLOGY BASED ON THE PHYSICAL CONNECTION OF COMPUTERS

SUGGESTED TIME: 7 PERIODS

IN THIS CHAPTER

This chapter explores how computers are physically interconnected to form networks: the main topologies, their strengths and weaknesses, and the criteria for choosing among them. It then moves from theory to practice by presenting network equipment and guiding the learner through the design and setting up of a complete organisational network.

- By the end of this chapter I can define a network topology and distinguish physical from logical topology.
- By the end of this chapter I can describe the structure and functioning of bus, star, ring, mesh and tree (hierarchical) topologies.
- By the end of this chapter I can compare topologies using criteria such as cost, reliability, scalability and ease of troubleshooting.
- By the end of this chapter I can justify the choice of a topology for a given real-life situation (school, cybercafé, bank, ministry).
- By the end of this chapter I can identify and state the role of common network equipment: NIC, hub, switch, router, modem, access point, repeater, cables and connectors.
- By the end of this chapter I can select appropriate equipment and cabling for a small organisational network given a budget and requirements.

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METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

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1. Physical Network Topologies
2. Comparing and Choosing a Topology
3. Network Equipment
4. Setting Up a Computer Network
5. Selecting Equipment for Organisational Networks
6. Integration activity
7. Methods and worked examples
8. Exercises
9. Solutions to the exercises
10. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can define a network topology and distinguish physical from logical topology.
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- By the end of this chapter I can select appropriate equipment and cabling for a small organisational network given a budget and requirements.
- By the end of this chapter I can plan and carry out the physical setting up of a simple peer-to-peer or client-server network.
- By the end of this chapter I can test, troubleshoot and document a small network installation.

PREREQUISITES

- Definition, advantages and types of computer networks (LAN, MAN, WAN) seen in

Lower Sixth

- Basic computer hardware components (system unit, ports, peripherals)
- Transmission media basics: guided vs unguided media (Lower Sixth introduction to networks)
- Notion of data transmission, bandwidth and simple units (bit/s, Mb/s)
- Elementary electricity and safety precautions when handling equipment

1 PHYSICAL NETWORK TOPOLOGIES

At a busy cybercafé in Buea, the owner complains that whenever one cable is accidentally unplugged, the whole network goes down and customers lose their work. At a bank branch in Douala, however, a single faulty computer never interrupts the other tellers. Why do two networks built from similar computers behave so differently? The answer lies in how the machines are physically connected — the **network topology** — and in the equipment chosen to link them. In this section, you will learn to identify, analyse and compare the fundamental physical topologies so that you can later design, set up and troubleshoot networks suited to real Cameroonian institutions.

1.1 What Is a Network Topology?

DEFINITION **Network Topology**

The network topology is the arrangement (layout) of nodes (computers, printers, servers, etc.) and the connecting links (cables, wireless channels) in a computer network. It describes the geometric pattern formed by the interconnections.

DEFINITION **Physical vs. Logical Topology**

Physical topology refers to the actual layout of cables, devices and connection points — what you see when you walk into the server room or trace the wires under the desks.
Logical topology refers to the way data signals actually travel through the network from source to destination, regardless of the physical wiring.
Example: A modern office wired with twisted-pair cables to a central switch has a physical star topology. If that switch is a hub (legacy), data is broadcast to every port → logical bus. If the switch is a modern switch, data is forwarded only to the destination port → logical star (point-to-point).

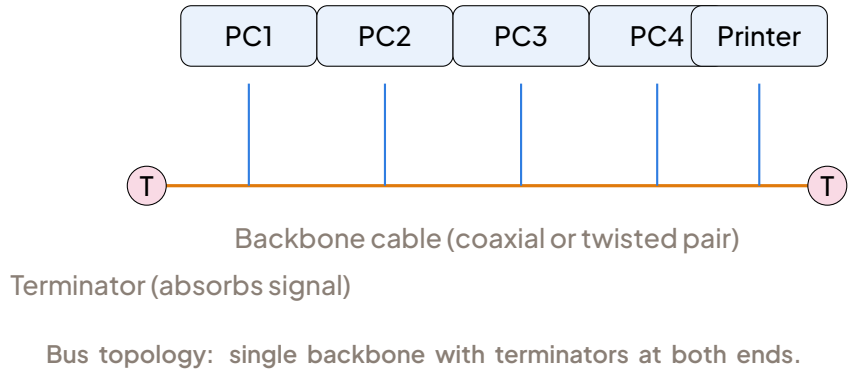
WARNING **Common Confusion**

Do not confuse the physical shape with the logical behaviour. A network that looks like a star (all cables meeting at a central box) may behave like a bus if the central box is a hub. Always check the equipment (hub vs. switch) to decide the logical topology.

1.2 Bus Topology

In a **bus topology**, all nodes connect to a single central cable — the **backbone** or **bus** — via drop lines and taps. Signals travel in both directions along the backbone. Each end of the cable must be fitted with a **terminator** (a resistor matching the cable's characteristic impedance) to ab-

sorb signals and prevent reflections that would corrupt data.



Bus topology: single backbone with terminators at both ends.

How data moves: When a node transmits, the signal propagates in both directions along the backbone. Every node receives the frame, but only the intended destination (matched by MAC address) processes it; others discard it. This is a **broadcast** medium — only one node can effectively transmit at a time (half-duplex), typically managed by CSMA/CD (Carrier Sense Multiple Access with Collision Detection).

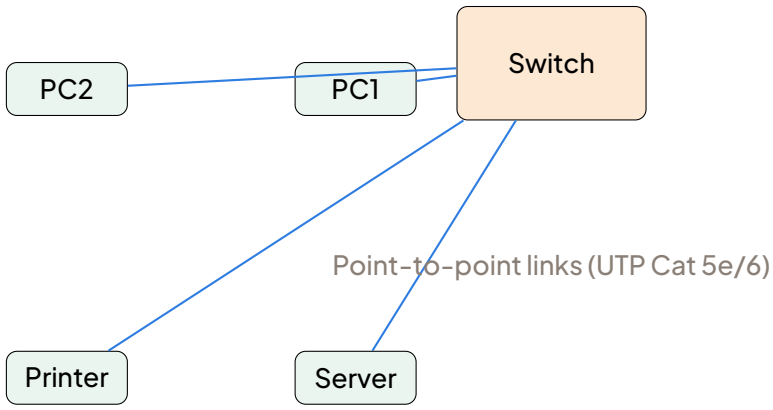
Effect of a break: A single cable break anywhere on the backbone splits the network into two isolated segments. **All** communication stops. This is the cybercafé scenario in our hook.

KEY POINTS **Bus Topology — Key Points**

- Simple, cheap cabling (less cable than star).
- Easy to extend by adding a tap.
- Single point of failure: backbone break kills the whole network.
- Difficult to isolate faults; performance drops as more nodes contend for the bus.
- Legacy: 10BASE2 (thinnet), 10BASE5 (thicknet). Rarely used for new installations.

1.3 Star Topology

In a **star topology**, every node connects via a dedicated point-to-point link to a central device — historically a **hub**, today almost always a **switch**. The central device acts as a traffic controller.



Star topology: all nodes connect to a central switch.

Failure modes:

- **One link / one node fails:** Only that node is isolated. The rest of the network continues unaffected (bank teller scenario).
- **Central switch fails: Entire network goes down.** This makes the switch a critical single point of failure. Redundancy (dual power supplies, stacking, or a backup switch) is essential in production networks.

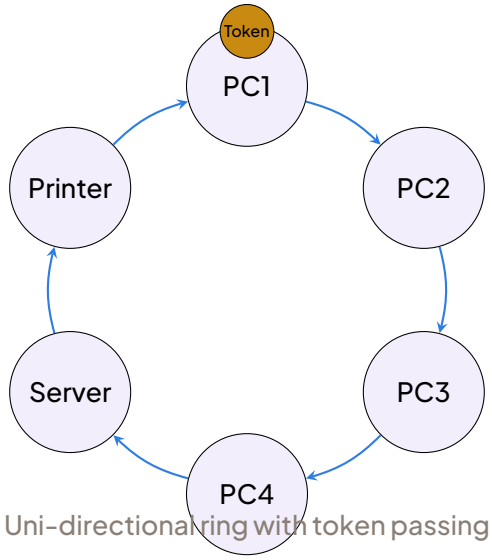
WARNING **Hub vs. Switch — Physical Star, Different Logical Behaviour**

- Hub (Layer 1 repeater): Electrically repeats every incoming bit out of all other ports. Logical topology = bus. Collisions propagate everywhere. Half-duplex only.
- Switch (Layer 2 bridge): Learns MAC addresses, forwards frames only to the destination port. Logical topology = star (point-to-point). Full-duplex possible. No collisions between different ports.

Exam tip: If a question says "star topology" without specifying the central device, assume a modern switch. If it says "hub", explicitly state the logical bus behaviour.

1.4 Ring Topology

In a **ring topology**, each node connects to exactly two neighbours, forming a closed loop. Data travels in one direction (uni-directional ring) or both directions (bi-directional / dual ring). Access is controlled by a **token** — a special control frame that circulates when the network is idle. A node may transmit only when it captures the token.



Ring topology: token circulates; only token holder transmits.

Single vs. Dual Ring:

- **Single ring:** One cable break breaks the whole ring. Used in Token Ring (IEEE 802.5) and FDDI (single attachment stations).
- **Dual ring (FDDI, some resilient Ethernet rings):** Two counter-rotating rings. On a break, the ring wraps at the neighbours of the fault, restoring a single logical ring. High availability.

KEY POINTS **Ring Topology — Key Points**

- Deterministic access (token) → predictable latency, good for real-time.
- No collisions.
- Adding/removing nodes disrupts the ring (unless using a MAU — Multistation Access Unit — which bypasses failed ports).
- Largely replaced by switched Ethernet, but concepts appear in resilient metropolitan rings (e.g., CAMTEL backbone rings).

1.5 Mesh Topology

In a **mesh topology**, nodes are interconnected with multiple redundant paths. We distinguish:

- **Full mesh:** Every node connects directly to every other node.
- **Partial mesh:** Only some pairs have direct links; others route through intermediate nodes.

PROPERTY **Number of Links in a Full Mesh**

For n nodes, a full mesh requires exactly

$$L = \frac{n(n-1)}{2}$$

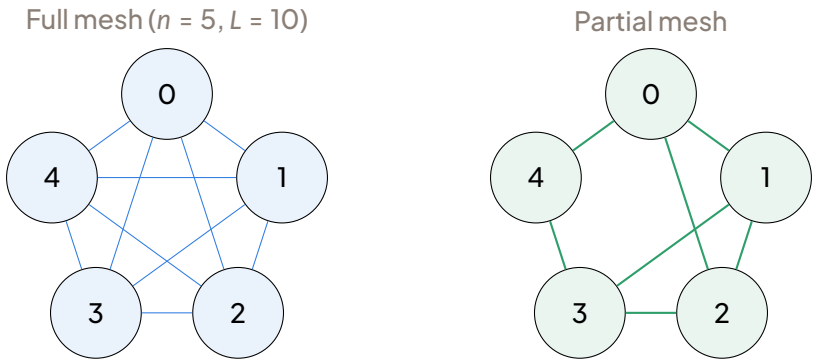
dedicated links. This grows quadratically ($O(n^2)$), making full mesh practical only for small n (e.g., core router interconnects, small critical clusters).

EXEMPLE **Calculating Mesh Links**

A ministry department in Yaoundé wants a full mesh among 6 servers for maximum redundancy. How many direct links are needed?

$$L = \frac{6 \times 5}{2} = 15 \text{ links.}$$

If they add a 7th server, $L = 21$ links — 6 new cables just for that one server. This is why full mesh is reserved for the core layer of large networks, not the access layer.



Full mesh vs. partial mesh (5 nodes).

Redundancy & Routing: Multiple paths allow automatic failover (via routing protocols like OSPF, or spanning tree at Layer 2). Partial mesh is the standard for WANs and enterprise core networks (e.g., MTN Cameroon backbone).

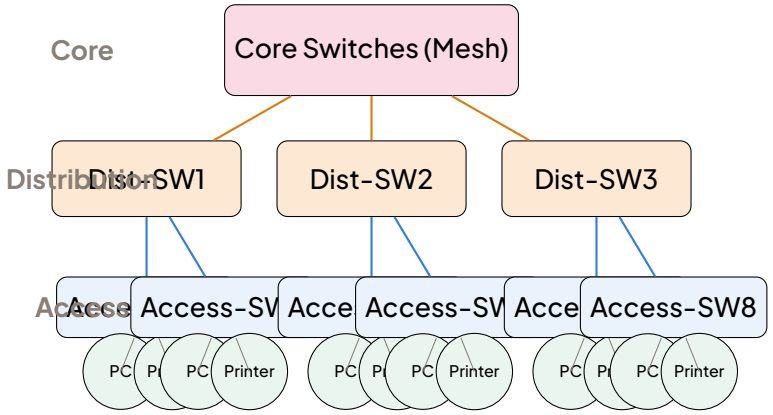
1.6 Tree (Hierarchical) Topology

A **tree topology** (or **hierarchical topology**) combines multiple star networks by connecting their central switches in a bus or higher-level star arrangement. It organises the network in **layers**:

- **Core layer:** High-speed backbone switches (often partial mesh).
- **Distribution (aggregation) layer:** Switches that aggregate access-layer switches, enforce

policies, routing between VLANs.

- **Access layer:** Switches to which end devices (PCs, APs, printers) connect — pure star.



Three-layer hierarchical (tree) topology — standard for campus/enterprise LANs.

Use in Cameroon: University campuses (e.g., University of Buea, University of Yaoundé I), ministry buildings in Yaoundé, large secondary school computer labs. The hierarchy limits broadcast domains, simplifies troubleshooting, and allows modular growth.

1.7 Hybrid Topologies and Real-World Cameroonian Examples

A **hybrid topology** combines two or more basic topologies. In practice, almost every real network is hybrid. Recognising the dominant topology in each segment is the key skill.

EXPERIMENT **Identifying Topologies in Cameroonian Scenarios**

Scenario 1: School Computer Lab (30 PCs)
Observation: All PCs wired to a 48-port switch in a locked cabinet.
Physical topology: Star.
Logical topology: Star (switch forwards selectively).
Resilience: One cut cable affects one PC. Switch failure kills lab → keep a spare switch.
Scenario 2: Cybercafé in Buea (Hook Revisited)
Observation: Single coaxial cable (or daisy-chained hubs) running along the wall, PCs tapped off it.
Physical topology: Bus (or extended star with hubs → logical bus).
Failure: One unplugged cable / broken coax → whole café down.
Upgrade path: Rewire to star with a switch.
Scenario 3: Ministry LAN (Multi-building)
Observation: Each building has an access switch (star). Building switches connect to a distribution switch in the main block (star of stars = tree). Distribution switches interconnect in a partial mesh for redundancy.
Overall: Hybrid (Tree at access/distribution, Mesh at core).

1.8 Contrasting Physical and Logical Topology: A Visual Summary

The following diagram shows the same physical star wiring used with a hub (logical bus) versus a switch (logical star).



Same physical star cabling; logical topology depends on central device (Hub vs. Switch).

1.9 Method: Recognising a Topology from a Network Diagram

METHOD **Identify the Physical Topology in 3 Steps**

1. Locate the central point(s). Is there a single device (switch/hub) to which all nodes connect? → Star. Is there a single cable running past all nodes? → Bus. Do nodes form a closed loop? → Ring. Are there many interconnections? → Mesh.
2. Count the links per node.
   - 1 link (end device) + 1 central device with many ports → Star.
   - 2 links (daisy-chained) → Bus or Ring (check if loop closes).
   - $n - 1$ links → Full Mesh.
3. Check for hierarchy. If you see distinct layers (core → distribution → access), it is a Tree (hierarchical) topology, possibly with a meshed core.

WORKED EXAMPLE 1 **Topology Identification**

Statement: A network diagram shows 4 switches labelled SW-A, SW-B, SW-C, SW-D. SW-A connects to SW-B, SW-C and SW-D. SW-B, SW-C, SW-D each connect to 20 PCs. SW-B, SW-C, SW-D are not directly connected to each other. Identify the topology.
Solution:

1. Access layer: Each of SW-B, SW-C, SW-D forms a star with its 20 PCs.
2. Distribution/Core: SW-A connects to the three switches in a star pattern (SW-A is the centre).
3. Overall: A Tree (hierarchical) topology with two levels (core SW-A, access SW-B/C/D). No mesh, no ring, no bus.

1.10 Comparing and Choosing a Network Topology (Lesson 15)

Choosing a topology depends on the organisation's requirements: cost, scalability, fault tolerance, performance, and ease of maintenance. The table below summarises the trade-offs.

Criterion	Bus	Star	Ring	Mesh (Partial)
Cabling cost	Low (single backbone)	Moderate (dedicated runs)	Moderate	High (multiple links)
Fault tolerance	Very poor (backbone break = total outage)	Good (link/PC failure isolated); switch = single point of failure	Poor (single ring); Excellent (dual ring)	Excellent (multiple paths)
Scalability	Poor (performance drops with nodes)	Excellent (add ports/switches)	Poor (disrupts ring to add nodes)	Good (add links selectively)
Performance	Low (shared medium, collisions)	High (dedicated bandwidth per port, full-duplex)	Deterministic (token), moderate bandwidth	High (load balancing over paths)
Troubleshooting	Difficult (fault isolation hard)	Easy (test per link/port)	Moderate (need to locate break)	Complex (requires routing/spanning tree)
Typical use today	Legacy only	**Standard for LAN access**	Metropolitan fibre rings, industrial	Core/backbone, WAN, data centres

METHOD **Choosing a Topology — Decision Guide**

1. Small office / school lab (< 50 nodes, budget tight): Star with a single switch. Low cost, easy to manage, sufficient resilience.
2. Multi-floor building / campus: Tree (hierarchical) — access switches per

floor/building, distribution switches, meshed core.

3. High-availability server cluster (4–8 nodes): Full mesh (or dual-ring) for sub-millisecond failover.
4. ISP / Telco backbone (MTN, Orange, CAMTEL): Partial mesh with dynamic routing (OSPF/IS-IS) and MPLS.
5. Industrial control / real-time: Dual ring (or Ethernet ring protection switching — ERPS) for deterministic recovery < 50 ms.

WARNING **Exam Traps — Choosing Topology**

- "Cheapest for 10 PCs in one room" → Star (UTP + switch), not Bus (coaxial is obsolete, hard to source, no NICs support it).
- "No single point of failure" → Requires redundancy: dual core switches, dual uplinks, partial mesh. A simple star has a SPOF (the switch).
- "Deterministic latency for VoIP/video" → Switched star with QoS is standard today; token ring is legacy. Do not recommend ring for new LANs.

1.11 Integration Activities (Lesson 16)

EXPERIMENT **Guided Design: Upgrading a Cybercafé Network**

Context: A cybercafé in Limbé has 20 PCs connected via a single coaxial bus (10BASE2). The owner wants to upgrade to support 30 PCs, enable Internet sharing, and eliminate total outages when one cable fails.
Step 1 — Requirements analysis:

- 30 endpoints → need 48-port switch (or 2 × 24-port stacked).
- Internet gateway → router/firewall (e.g., MikroTik, Ubiquiti).
- Wireless for customers → 2–3 Access Points (APs) connected to switch.
- Power protection → UPS for switch, router, APs.

Step 2 — Topology selection: Star (wired) + Star (wireless via APs) = Hybrid Star. All wired runs terminate at a patch panel in a locked cabinet → switch.
Step 3 — Equipment list (Cameroon market typical):

- 1 × 48-port Gigabit managed switch (TP-Link TL-SG3428 / Cisco SG350 / Ubiquiti USW-48).

- 1 × Router/Firewall (MikroTik hEX / Ubiquiti EdgeRouter).
- 3 × Ceiling-mount APs (Ubiquiti U6-Lite / TP-Link EAP650).
- 30 × Cat 6 UTP cables (3 m drops), 1 × 24-port patch panel, 1 × 9U wall cabinet.
- 1 × 1.5 kVA UPS (APC / Mercury).

Step 4 — Implementation checklist:

1. Run all cables, label both ends (patch panel port ↔ wall jack).
2. Terminate on patch panel (T568B standard), test with cable certifier.
3. Connect patch panel to switch ports; connect router WAN to ISP modem, LAN to switch.
4. Configure switch: VLAN 10 (wired PCs), VLAN 20 (wireless guests), VLAN 99 (management).
5. Configure router: DHCP per VLAN, firewall rules (guest VLAN no access to admin VLAN), bandwidth limiting per user.
6. Configure APs: same SSID, WPA2-Enterprise (RADIUS) or captive portal for guests.
7. Document: network diagram, IP plan, VLAN table, switch port map.

Step 5 — Verification: Ping test from each PC, roaming test between APs, failover test (unplug one AP → clients roam), bandwidth test (Speedtest).

1.12 Network Equipment Overview (Lesson 17 — Introduction)

Understanding topology requires knowing the devices that implement it. This section introduces the key equipment; detailed configuration is covered in later chapters.

DEFINITION **Network Equipment Categories**

- End devices (Hosts): PCs, laptops, smartphones, printers, IP phones, servers — sources/destinations of data.
- Intermediary devices (Network devices): Forward, filter, or direct data traffic.
  - Hub (Layer 1): Multi-port repeater. Obsolete for LANs. Creates one collision domain.
  - Switch (Layer 2): MAC-address learning, forwards frames per port. Creates separate collision domains per port. Modern standard.
  - Wireless Access Point (AP, Layer 2): Bridges wireless (802.11) to wired Ethernet. Acts like a hub for RF medium (CSMA/CA).

- Router (Layer 3): Interconnects different networks (IP subnets), makes forwarding decisions based on IP address. Essential for Internet access and inter-VLAN routing.
- Firewall (Layer 3/4/7): Enforces security policies (allow/deny/inspect). Can be dedicated appliance or router feature.
- Modem / ONT: Demarcation device converting ISP signal (fibre GPON, xDSL, 4G/5G) to Ethernet.

- Media: UTP Cat 5e/6/6a (100 m max), Fibre optic (single-mode / multi-mode, km range), Wireless (2.4/5/6 GHz).

KEY POINTS **Equipment–Topology Mapping**

- Bus: Coaxial cable + BNC T-connectors + terminators (legacy). No active central device.
- Star: Central switch (wired) or AP (wireless).
- Ring: Each node has two interfaces; or MAU (Multistation Access Unit) for Token Ring; or fibre switches in ring with ERPS.
- Mesh: Routers or Layer 3 switches with multiple interfaces running dynamic routing (OSPF, BGP).
- Tree: Access switches (Layer 2) → Distribution switches (Layer 2/3) → Core switches (Layer 3, high throughput).

1.13 Summary of Physical Topologies

Topology	Key Physical Trait	Failure Impact	Typical Use in Cameroon
Bus	Single backbone, terminators	Backbone break = total outage	Legacy cybercafés, old school labs
Star	Central switch, point-to-point links	Link/PC fail = isolated; Switch fail = total outage	Modern labs, offices, SME LANs
Ring	Closed loop, token passing	Single ring: break = total outage; Dual ring: self-heals	Metropolitan fibre rings (CAMTEL), some industrial
Full Mesh	Every node ↔ every node ($n(n-1)/2$ links)	Highly resilient; multiple paths	Core router interconnects, critical server clusters
Partial Mesh	Selected redundant links	Good resilience, scalable	Enterprise core, ISP backbone (MTN, Orange)
Tree (Hierarchical)	Stars of stars in layers (Core/Dist/Access)	Localised failures; core failure major	University campuses, ministries, large schools

DID YOU KNOW? **Cameroon Context: ANTIC and Network Standards**

The Agence Nationale des Technologies de l'Information et de la Communication (ANTIC) promotes secure and resilient network architectures for public administration. Their guidelines for government LANs (e.g., in the Plan National de Développement du Numérique) recommend hierarchical tree topologies with a meshed core, VLAN segmentation, and redundant links — exactly the hybrid model studied here. When you design a network for a ministry or council, align with this model.

EXERCISE 1 **Practice: Topology Analysis**

1. A small NGO office in Bamenda has 8 laptops connected to a single 8-port hub. Draw the physical topology. What is the logical topology? What happens if the hub fails?
2. A university campus connects 5 faculty buildings. Each building has a star-wired LAN. The 5 building switches are connected in a ring using fibre. Name the overall topology. If the fibre between Building 2 and Building 3 is cut, can Building 1 still reach Building 4? Explain.
3. Calculate the number of direct links required for a full mesh among 12 core routers. Why is a partial mesh preferred for this scale?
4. Exam-style: "A network uses a central device to which all computers are connected

by individual cables. If one cable is cut, only that computer loses connectivity. However, if the central device fails, the whole network stops. Name the physical topology and the most likely central device used today." (3 marks)

5. Design question: A secondary school in Douala wants to network 3 computer labs (25 PCs each), an admin block (10 PCs), and a library (15 PCs). The labs are in separate buildings 50 m apart. The admin block is 100 m from the nearest lab. Propose a topology, justify your choice, and list the main equipment needed (switches, fibre/copper, router). (6 marks)

SOLUTION **Exercise 1**

1. Physical: Star. Logical: Bus (hub broadcasts). Hub failure → total outage.
2. Overall: Hybrid (Tree of Stars + Ring). Yes, traffic goes the other way around the ring (Building 1 → Building 5 → Building 4).
3. $L = 12 \times 11/2 = 66$ links. Partial mesh preferred because 66 cables/ports is expensive and hard to manage; routing protocols provide redundancy with fewer links.
4. Physical topology: Star. Central device: Switch (modern) or Hub (legacy). Today: Switch.
5. Proposed topology: Tree (hierarchical).
   Justification: Separate buildings → each building = access layer (star). Building switches uplink to a distribution switch in admin block (or central lab). Fibre for inter-building links (distance > 100 m, immunity to lightning). Copper (Cat 6) inside buildings. Router at distribution for Internet/VLAN routing.
   Equipment:
   - 3 × 24-port PoE+ access switches (labs + library), 1 × 12-port switch (admin).
   - 1 × 24-port Layer 3 distribution switch (core) with SFP+ uplinks.
   - 4 × SFP+ modules + 4 × OM3 multi-mode fibre patch cords (inter-building).
   - Cat 6 UTP for all horizontal runs (patch panels, wall jacks).
   - 1 × Router/Firewall (Internet gateway, VLAN routing).
   - 1 × UPS for distribution switch + router.

2 COMPARING AND CHOOSING A TOPOLOGY

In the previous section we described the five main **physical topologies** — bus, star, ring, mesh

and tree — and how computers are physically connected in each of them. Knowing the shapes, however, is not enough. A network engineer in Douala or Bamenda who is asked to network a school computer laboratory, a bank branch or a hospital must answer a harder question: which topology should we actually install, and why? This section gives you the comparison criteria, a detailed comparison of the five topologies, and a rigorous method for justifying a choice — exactly what GCE Advanced Level examiners expect.

2.1 Comparison Criteria

DEFINITION **Comparison criteria for network topologies**

A **comparison criterion** is a measurable or observable property used to evaluate and rank topologies. The five standard criteria are: cost (cabling and equipment), reliability / fault tolerance (what happens when a cable or device fails), performance (speed, collisions, delays), scalability (ease of adding new stations), and ease of installation and maintenance (including fault detection).

Let us build the intuition behind each criterion before comparing.

Cost. Cabling is often the largest hidden expense of a network. A bus uses one single backbone cable, so it is cheap in cable length; a full mesh with n computers needs $\frac{n(n-1)}{2}$ separate links, which explodes quickly. Equipment cost also matters: a star needs a central switch, while a bus needs only terminators.

Reliability / fault tolerance. Ask: if one cable is cut, does the whole network stop? In a bus, cutting the backbone kills everything. In a star, cutting one branch cable disconnects only one computer — but if the switch fails, everything stops (single point of failure). In a mesh, traffic simply takes another path.

Performance and collisions. On a shared medium (bus, or a ring without proper token control), two stations transmitting simultaneously cause a **collision**, and both must retransmit. The more stations, the more collisions, and the slower the effective throughput. A switched star gives each computer a dedicated link to the switch, so collisions are practically eliminated.

Scalability. Adding a computer to a star means plugging one cable into a free switch port — seconds of work. Adding one to a bus may require shutting the network down and extending the backbone; adding one to a full mesh requires a new cable to every existing machine.

Ease of installation and maintenance. Centralised topologies (star, tree) make fault-finding easy because all connections meet at one point. Distributed topologies (bus, mesh) force the technician to hunt along walls and ceilings for the faulty segment.

2.2 Detailed Comparison of the Five Topologies

Criterion	Bus	Star	Ring	Mesh	Tree
Cost of cabling & equipment	Very low: one backbone, no central device	Moderate: one cable per host + switch	Moderate: ring cable + MAU	Very high: $\frac{n(n-1)}{2}$ links, many ports	High: hierarchical cabling + several switches
Reliability / fault tolerance	Poor: backbone break stops all	Good per link; switch is single point of failure	Poor: one break stops ring (unless dual ring)	Excellent: multiple alternative paths	Good: failure isolated to one branch; root switch critical
Performance / collisions	Degrades fast: shared medium, collisions	High: dedicated links, no collisions with switches	Fair: token passing gives orderly access, delay grows with n	Very high: direct dedicated links	High within branches; backbone can bottleneck
Scalability	Poor: adding hosts is disruptive	Excellent: plug into a free port	Poor: ring must be opened	Very poor: new links to all nodes	Excellent: add a branch or a switch
Installation & maintenance	Simple to install, hard to troubleshoot	Easy to install; faults found centrally	Moderate; fault location tricky	Very complex to install and manage	Moderate; structured, manageable per branch

Comparison of the five physical topologies across the five standard criteria.

KEY POINTS **Advantages and disadvantages of each topology**

Bus — Advantages: cheapest cabling, simple, uses little cable. Disadvantages: backbone failure stops everything, collisions, hard fault-finding, limited size.
Star — Advantages: easy to install, extend and troubleshoot; a cable failure affects only one host; excellent performance with switches. Disadvantages: needs a central device (cost); switch failure stops the whole network; more cable than a bus.

Ring — Advantages: orderly access (token), no collisions, predictable delay. Disadvantages: one break can stop the ring, adding/removing hosts is disruptive, largely obsolete.
Mesh — Advantages: outstanding fault tolerance, dedicated high-speed links, secure point-to-point paths. Disadvantages: extremely expensive in cable and ports, complex to install and configure.
Tree — Advantages: highly scalable, groups departments naturally, faults isolated per branch. Disadvantages: depends on backbone and root switch, more cabling and equipment than a simple star.

PROPERTY **Dominance of the star topology**

For the great majority of local area networks, the **star topology** offers the best balance of cost, reliability and scalability: a single cable failure isolates only one machine, new stations are added in seconds, and modern switches eliminate collisions at moderate cost. This is why the star (and its extension, the tree) dominates virtually all modern Ethernet LANs. (Justification discussed in class; not a formal proof to reproduce in the exam.)

2.3 Decision Factors in Context

Choosing a topology is never done in the abstract. The correct answer always depends on the scenario:

- **Size of the organisation.** A handful of machines can live with almost anything; hundreds of machines demand the structure and scalability of a star or tree.
- **Budget.** A cybercafé in Buea with limited capital will avoid mesh; a bank can afford redundancy.
- **Expected growth.** If the organisation plans to double its computers in two years, scalability (star/tree) outweighs initial savings (bus).
- **Criticality of service.** In a hospital, where a network failure can endanger lives, or in a bank processing mobile-money transactions, fault tolerance dominates the decision. In a small cybercafé, an hour of downtime is annoying but not catastrophic, so cost may dominate.

WARNING **The mesh trap**

A very common mistake — in the exam and in real life — is to answer "mesh" whenever reliability is mentioned, without considering its cabling cost and complexity. With n computers, a full mesh needs $\frac{n(n-1)}{2}$ links: 30 computers already require 435 cables! Mesh is justified only for a small number of critical interconnection points (e.g. linking backbone routers). For end-user machines, a star with a backup switch usually gives sufficient reli-

ability at a fraction of the cost.

2.4 Case Analyses

EXEMPLE **Case 1 — A 20-computer school laboratory**

Needs: low budget, one room, easy maintenance by one teacher, moderate reliability. Choice: star. One 24-port switch at the teacher's desk, one cable per computer. A faulty cable disconnects only one student machine; adding computers later means plugging into free ports. A bus would be cheaper in cable but a single backbone fault would stop the whole class — unacceptable during a practical exam session.

EXEMPLE **Case 2 — A bank branch**

Needs: high availability (transactions must never stop), security, moderate number of stations. Choice: star with redundancy — a star per floor or department, with a backup switch and dual links to the branch server; the interconnection between the critical core devices may use a partial mesh. Pure mesh for every teller machine would be needlessly expensive; a bus or ring would risk total outage.

EXEMPLE **Case 3 — Connecting three buildings of a campus**

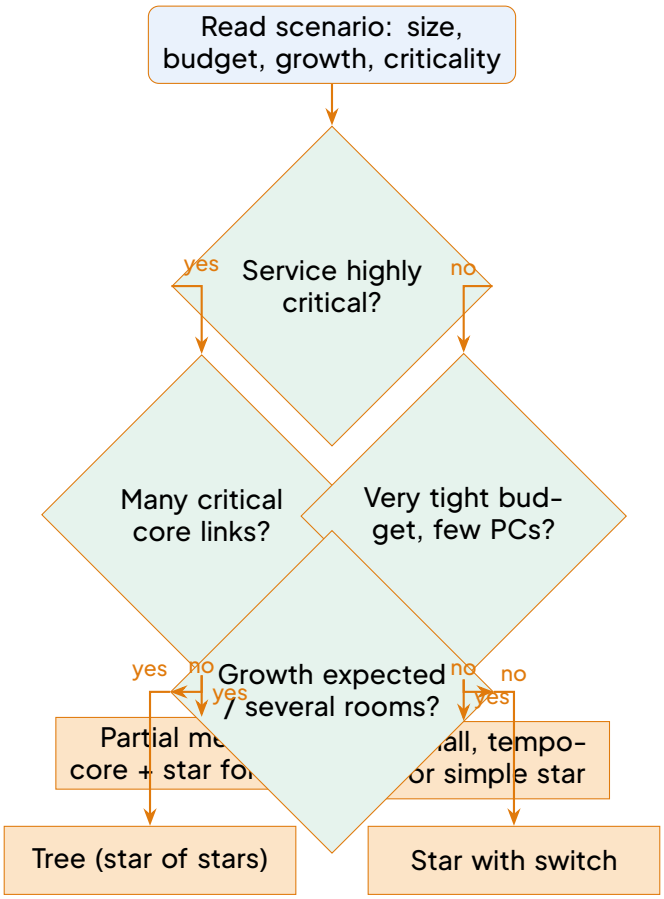
Needs: each building has its own LAN; buildings must communicate; the network must grow. Choice: tree (hierarchical star-of-stars). Each building keeps an internal star with its own switch; the three building switches connect to a central backbone switch (often by fibre optic). A fault in one building does not affect the others, and a fourth building can be added later by connecting one more branch.

2.5 A Rigorous Method for Justifying a Choice

METHOD **Four-step procedure to justify a topology choice**

1. Identify the needs from the scenario: number of stations, budget, expected growth, criticality of service, physical layout (one room? several buildings?).
2. List the candidate topologies that could physically fit (usually two or three of bus, star, ring, mesh, tree).

3. Compare the candidates on at least three criteria (cost, reliability, performance, scalability, maintenance), linking each criterion explicitly to the scenario.
4. Conclude and justify: state the chosen topology and explain why it beats the rejected candidates on the decisive criteria, acknowledging its disadvantages and how they are mitigated.



Decision flowchart: from requirements to a justified topology choice.

WORKED EXAMPLE 2 **GCE-style question**

A new regional hospital in Bamenda has 45 computers spread over three wards. The network carries patient records and must remain available at all times. The hospital expects to open a fourth ward next year. Recommend a physical topology, justifying your answer with at least three criteria.

SOLUTION

Step 1 — Needs: 45 stations, several buildings/wards, high criticality (patient records, 24/7 availability), planned growth (a fourth ward). Budget is secondary to reliability.
Step 2 — Candidates: star (per ward), tree (campus-wide), mesh (for core links). Bus and ring are rejected immediately: a single break would stop a ward.
Step 3 — Comparison on criteria:

- Reliability: a tree isolates a fault to one ward, so the other wards keep working — essential for a hospital. A single flat star would make the whole hospital depend on one switch.
- Scalability: the fourth ward is added next year by connecting one new branch switch to the backbone — a tree handles this naturally; a bus or ring would require disruptive rewiring.
- Performance: each ward's star gives dedicated switched links (no collisions), and traffic between wards uses the backbone only when needed.
- Cost: a full mesh for 45 machines would need $\frac{45\times44}{2}$ = 990 links — absurdly expensive; the tree achieves the required reliability at a fraction of that cost.

Step 4 — Conclusion: we recommend a tree topology (a star per ward, ward switches linked to a central backbone switch, with a backup backbone link for extra fault tolerance). It best satisfies reliability, scalability and performance for this critical, growing hospital network, at a justifiable cost.

WARNING **Exam tip — link criteria to the scenario**

In GCE questions, marks are awarded for argued criteria, not for reciting definitions. Never write "star is reliable" in isolation. Write: "a star is chosen because in this laboratory a cable fault must not stop the other 19 students during practical examinations." Every criterion you cite must be tied to a fact given in the scenario (size, budget, growth, criticality). A correct topology with no scenario-linked justification earns only partial credit.

EXERCISE 2 **Choosing and justifying**

A cybercafé in Limbe has 8 computers in one small room and a very limited budget; the owner may add 4 more computers next year. A microfinance bank in Yaoundé has 25 workstations and must guarantee continuous service for mobile-money operations. For each organisation, recommend a topology and justify it with at least three criteria linked to the scenario.

SOLUTION **Exercise 1**

Cybercafé: star with a small 16-port switch. Criteria: (i) cost — one inexpensive switch and short cables suit the limited budget; (ii) scalability — the 4 future computers plug into the free ports, so growth is effortless; (iii) maintenance — the owner can identify a faulty machine instantly at the switch, important since there is no paid technician. A bus would be slightly cheaper but a backbone fault would close the whole café.
Microfinance bank: star with redundancy (backup switch, dual link to the server), possibly partial mesh between the core devices. Criteria: (i) reliability — continuous service for mobile-money transactions is critical, so no single cable or switch failure may stop operations; (ii) performance — switched dedicated links avoid collisions and keep transaction times short; (iii) fault isolation — a faulty teller cable disconnects only that teller. Full mesh for all 25 stations (300 links) is rejected as unnecessarily costly.

3 NETWORK EQUIPMENT

In the previous section we compared the physical topologies (bus, ring, star, mesh) and concluded that the **star topology** is the standard choice for modern local area networks. But a topology is only a plan on paper: to transform that plan into a working network, we need concrete **network equipment** — the cards, cables, connectors and devices that physically carry the data. In this section we study each piece of equipment in turn: what it is, what it does, and where it is used. At the GCE Advanced Level, examiners frequently present a network diagram and ask: "State the role of device X in this network." By the end of this section you will be able to answer such questions precisely.

3.1 The Network Interface Card (NIC)

DEFINITION **Network Interface Card**

A **Network Interface Card** (**NIC**) is the hardware component that connects a computer (or printer, or any other device) to a network. It prepares the computer's data for transmission over the medium and controls the flow of data between the computer and the cable or radio channel.

Intuition. Think of the NIC as the "mouth and ears" of the computer on the network. Without it, a computer simply cannot speak to or hear other machines. On modern motherboards the NIC is usually built in (onboard); on older machines it is a separate expansion card (PCI, PCIe, or USB).

Every NIC manufactured in the world carries a unique physical address called the **MAC address** (Media Access Control address). It is a 48-bit number, usually written as six pairs of hexadecimal digits, for example `3C:5A:B4:0F:2E:91`. The first three octets identify the manufacturer (OUI), the last three are a unique serial number. The MAC address is used by switches to identify ex-

actly which device is attached to which port.

There are two main variants:

- **Wired NIC:** has an RJ45 port for an Ethernet cable; typical speeds are 100 Mbit/s (Fast Ethernet), 1 Gbit/s (Gigabit Ethernet), or 2.5/10 Gbit/s on newer hardware.
- **Wireless NIC (Wi-Fi card):** has an antenna and communicates by radio waves with a wireless access point, following the IEEE 802.11 standards (Wi-Fi 4/5/6/6E/7).

WARNING **NIC drivers and binding**

A NIC requires a device driver in the operating system. In the GCE practical, you may be asked to verify that the NIC is recognised (e.g. `ipconfig /all` on Windows or `ip link` on Linux) and that it has a valid MAC address.

3.2 Transmission Media and Connectors

Data must travel on a physical medium. The three media you must know for the examination are the following.

Twisted pair cable. This is the most common LAN cable. It contains four pairs of copper wires twisted together to reduce electromagnetic interference (EMI) and crosstalk. Two versions exist:

- **UTP** (Unshielded Twisted Pair): no metallic shield; cheap, flexible, easy to install; used in offices, schools and homes.
- **STP** (Shielded Twisted Pair): each pair or the whole cable is wrapped in a metallic foil/braid; more resistant to interference; used in factories, near heavy electrical equipment, or for outdoor runs.

Twisted pair cables are terminated with an **RJ45 connector** (8P8C) and are classified into **categories**:

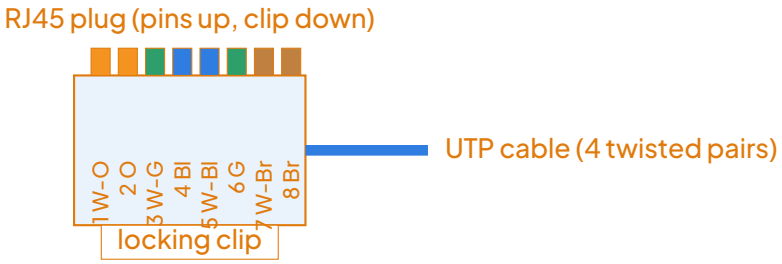
Category	Max Data Rate (typical)	Max Distance
Cat5e	1 Gbit/s	100 m
Cat6	1 Gbit/s (100 m), 10 Gbit/s (up to 55 m)	100 m
Cat6a	10 Gbit/s	100 m
Cat7 / Cat7a	10–40 Gbit/s	100 m

The maximum recommended length of a single twisted pair segment (horizontal cabling) is **100 m** (90 m solid cable + 10 m patch cords).

WARNING **Straight vs crossover cables**

A **straight-through cable** (T568B at both ends) has identical wire order at both ends and

connects different types of devices (PC to switch, switch to router). A **crossover cable** (T568A at one end, T568B at the other) swaps the transmit and receive pairs (pins 1&2 with 3&6) and connects similar devices directly (PC to PC, switch to switch on old equipment without auto-MDIX). Modern Gigabit interfaces implement auto-MDIX and automatically detect the cable type, making crossover cables rarely needed today. Confusing the two is a classic practical-exam mistake.



RJ45 connector pinout (T568B standard) viewed with the locking clip facing down. Pins 1 & 2 = transmit pair (orange), pins 3 & 6 = receive pair (green).

Coaxial cable. A central copper conductor surrounded by insulation, a metallic braid (shield) and an outer jacket. It was used in old bus networks (10BASE2 “thinnet”, 10BASE5 “thicknet”) and is still used for cable television (CATV) and some broadband Internet (DOCSIS). It is largely obsolete in LANs but you may encounter it in legacy installations.

Fibre optic cable. Transmits data as pulses of light through a glass or plastic core. It offers very high speeds, immunity to electromagnetic interference and distances of several kilometres. Two main types:

- **Single-mode fibre (SMF):** small core (8–10 µm), laser source, distances up to 100+ km; used for ISP backbones (CAMTEL, MTN fibre links between towns).
- **Multi-mode fibre (MMF):** larger core (50/62.5 µm), LED/VCSEL source, distances up to 550 m (10 Gbit/s); used for campus backbones, data centres.

Common connectors: SC, LC, ST, FC. In Cameroon, FTTH (Fibre To The Home) deployments by CAMTEL, MTN and Orange typically use an **ONT** (Optical Network Terminal) that converts light to Ethernet.

3.3 The Hub

DEFINITION **Hub**

A **hub** (multiport repeater) is a Layer 1 device that connects several computers in a physical star topology. When it receives bits on one port, it simply regenerates and broadcasts

those bits to all its other ports, regardless of the destination.

Intuition. A hub behaves like a town crier in the market square: it shouts every message to everybody, and each person must decide whether the message concerns them. This creates two serious limitations:

- **Collisions:** all ports share a single **collision domain**; if two computers transmit at the same time, the signals collide and must be re-sent (CSMA/CD), wasting time.
- **Wasted bandwidth and poor security:** every machine receives traffic meant for others (promiscuous mode can capture all frames).

For these reasons the hub is now **obsolete** and has been replaced by the switch. It survives in the syllabus mainly so that you can contrast it with the switch.

3.4 The Switch

DEFINITION **Switch**

A **switch** (multiport bridge) is a Layer 2 device that connects several devices in a star topology and forwards data frames only to the port of the destination device, using the destination **MAC address** contained in each frame.

How it works. The switch maintains a MAC address table (CAM table) that records which MAC address is reachable through which port. When a frame arrives, the switch reads the destination MAC address, consults its table, and sends the frame out of the correct port only (unicast). If the address is not yet known (unknown unicast), the switch temporarily floods the frame to all ports except the incoming one, then learns the sender's location from the reply. Each port is its own collision domain (full-duplex operation eliminates collisions entirely).

PROPERTY **Switch vs Hub – examinable comparison**

- Hub: Layer 1, one collision domain, half-duplex, broadcasts to all ports, 10/100 Mbit/s shared bandwidth.
- Switch: Layer 2, one collision domain per port, full-duplex, forwards/floods based on MAC table, dedicated bandwidth per port (100 Mbit/s, 1 Gbit/s, 10 Gbit/s).

WARNING **Hub ≠ switch**

A very common mistake is to treat a hub and a switch as identical. They are not: a hub broadcasts to all ports; a switch selects the destination port. The switch therefore re-

duces collisions, increases effective speed and improves confidentiality.

Managed vs Unmanaged. An unmanaged switch works out of the box (plug-and-play). A managed switch offers VLANs, port mirroring, QoS, SNMP, link aggregation (LACP), and can be configured via CLI/Web/SSH. School labs and enterprises use managed switches; homes and small cybercafés often use unmanaged.

The switch is the central device of the modern star topology studied in the previous sections: every computer in the lab or office is wired back to it.

3.5 The Router

DEFINITION **Router**

A **router** is a Layer 3 device that interconnects different networks and routes data packets between them, choosing the best path towards the destination using **IP addresses**. The router that links a local network to the outside world is called the **default gateway**.

Intuition. If the switch is the internal postman of one town (one network), the router is the sorting office that dispatches mail between towns (between networks). When a student in Buea opens a website hosted abroad, the packet leaves the school LAN through the router (default gateway), which forwards it towards the Internet.

The **home router** (often called "box" or "gateway") supplied by Internet providers (MTN, Orange, CAMTEL) is actually several devices in one box:

- a **router** (WAN port, NAT, firewall, DHCP server, DNS proxy);
- a small **switch** (typically 4 LAN ports);
- a **wireless access point** (Wi-Fi 5/6);
- often a **modem** (ADSL, VDSL, or fibre ONT integrated or separate).

KEY POINTS **Router functions in a SOHO device**

1. Inter-network routing: forwards packets between LAN and WAN using IP addresses.
2. NAT (Network Address Translation): allows multiple private IPs (e.g. 192.168.1.x) to share one public IP.
3. DHCP server: automatically assigns IP configuration to LAN clients.
4. Firewall / SPI: blocks unsolicited incoming traffic from the Internet.
5. Wireless AP: provides Wi-Fi access (see below).

3.6 The Modem

DEFINITION **Modem**

A **modem** (MOdulator–DEModulator) is a device that converts the digital signals of a computer network into signals suitable for a transmission line (telephone line, coaxial cable, fibre) and back again. It connects a LAN to an Internet Service Provider (ISP).

- **ADSL/VDSL modem:** uses the ordinary copper telephone line (PSTN); typical of older home connections; asymmetric speeds (download > upload).
- **Cable modem (DOCSIS):** uses the coaxial TV cable network; higher speeds.
- **Fibre modem / ONT (Optical Network Terminal):** converts light pulses from a fibre optic line (GPON, XGS-PON) into Ethernet signals; used with fibre-to-the-home offers in Douala, Yaoundé and major towns.

In modern FTTH installations, the ISP provides an ONT (modem) and a separate router, or a combined gateway.

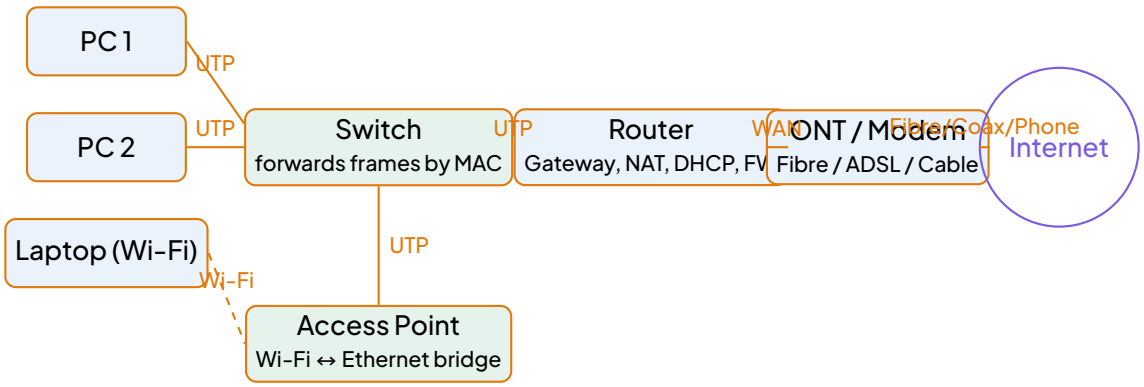
3.7 Wireless Access Point, Repeater and Bridge

- **Wireless Access Point (WAP / AP):** allows Wi-Fi devices (laptops, phones, tablets) to join a wired network. It is the wireless equivalent of a switch port. It bridges 802.11 (Wi-Fi) frames to 802.3 (Ethernet) frames. Multiple APs with the same SSID form an Extended Service Set (ESS) enabling roaming.
- **Repeater / Range extender:** receives a Wi-Fi signal and retransmits it to extend coverage, for example to carry Wi-Fi to the far end of a school building. It halves the available bandwidth (half-duplex relay). Modern mesh systems use dedicated backhaul radios to avoid this penalty.
- **Bridge:** connects two similar network segments (usually wired) so that they behave as one Layer 2 network, for example linking the wired LAN of two nearby classrooms or buildings via a wireless bridge (point-to-point link).

3.8 Other Installation Hardware (Structured Cabling)

- **Patch panel:** a panel of RJ45 sockets (usually 24 or 48 ports), mounted in a 19-inch cabinet/rack, where all the horizontal cables of a building terminate; short patch cords then link it to the switch. Provides flexibility and protects solid-core cable from repeated plugging.
- **Wall socket (keystone jack / faceplate):** the RJ45 outlet fixed in the wall or floor box, into which the user plugs a computer via a patch cord.
- **Crimping tool:** the pliers used to fix an RJ45 connector onto the wires of a patch cable (stranded wire). Not for punching down solid cable onto a patch panel or keystone (use a punch-down tool / Krone tool).

- **Cable tester:** a small device (main unit + remote) that checks whether all eight wires of a cable are correctly connected, detects opens, shorts, crossed pairs, and split pairs. Essential for practical exams.
- **PoE (Power over Ethernet) injector / PoE switch:** delivers DC power (up to 90 W with 802.3bt) over the same Ethernet cable to devices like APs, IP cameras, VoIP phones — no separate power adapter needed.



A small organisational LAN: PCs connect to a switch via UTP, the switch to a router (gateway), the router to an ONT/modem, and the modem to the ISP network. Wi-Fi clients join via an Access Point wired to the switch.

METHOD **Distinguishing hub, switch and router in exam questions**

Ask one question: "Who receives the data?"

1. If the device sends the data to everybody on the local network (Layer 1 broadcast) ⇒ it is a hub.
2. If it sends the data only to the one destination machine inside the same network (Layer 2, MAC address) ⇒ it is a switch.
3. If it sends the data out of the local network towards another network or the Internet (Layer 3, IP address) ⇒ it is a router.

WORKED EXAMPLE 3 **Identifying devices in a school network**

The computer laboratory of GBHS Bamenda has 20 PCs cabled to a central device which sends each frame only to the intended PC. This device is connected to a second device which gives access to the Internet through the school's CAMTEL fibre line. (a) Name the two devices. (b) State the role of each. (c) Why is the first device preferred to a hub?

SOLUTION

(a) The first device is a switch; the second is a router (acting as default gateway, with a fibre ONT towards CAMTEL).
(b) The switch connects the 20 PCs in a star topology and forwards each frame only to the port of the destination PC, using MAC addresses. The router interconnects the school LAN with the Internet and routes packets between the two networks using IP addresses; it also performs NAT so that all 20 PCs share the school's single public IP.
(c) A hub would broadcast every frame to all 20 PCs, causing collisions, wasting bandwidth and letting every PC see traffic meant for others. The switch selects only the destination port, so communication is faster, more efficient and more private.

EXPERIMENT **Guided practical: Build and test a patch cable**

Objective: Terminate a Cat6 UTP cable with two RJ45 plugs (T568B) and verify continuity. Equipment: Cat6 UTP cable (1 m), 2 RJ45 plugs, crimping tool, cable tester, wire stripper/cutter. Steps:

1. Strip ≈ 25 mm of outer jacket; untwist pairs only as needed.
2. Arrange wires in T568B order: White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.
3. Trim wires flush so ≈ 12 mm protrude; insert fully into RJ45 plug (each wire in its channel).
4. Crimp firmly; repeat for the other end (same order for straight-through).
5. Test: plug each end into the tester; all 8 LEDs should light in sequence. If any LED fails, re-crimp.

Exam tip: In the practical, you may be asked to identify a faulty cable (split pair, reversed pair, open circuit) using the tester.

EXERCISE 3 **Roles of network equipment**

For each situation, name the most appropriate device and justify your choice:

1. Connecting 30 desktop computers of a cybercafé in Douala into one LAN.
2. Giving Wi-Fi access to students' phones in that LAN.
3. Linking the LAN to the Internet via a CAMTEL fibre subscription.
4. Extending Wi-Fi coverage to a room 80 m away from the main router.

5. Interconnecting the wired LANs of two buildings 200 m apart using a wireless link.

SOLUTION **Exercise 1**

1. A switch (preferably managed, 48-port Gigabit): it connects the PCs in a star and forwards frames only to the destination port, providing dedicated bandwidth.
2. A wireless access point (PoE-powered, ceiling-mounted): it lets Wi-Fi devices join the wired LAN; multiple APs with same SSID allow roaming.
3. A router (with a separate ONT provided by CAMTEL): the ONT converts fibre to Ethernet; the router performs NAT, DHCP, firewall and connects the LAN to the ISP.
4. A Wi-Fi repeater/extender (or better, a second AP wired via UTP if distance ≤ 100 m, or a mesh system): it regenerates the Wi-Fi signal to cover the distant room.
5. A wireless bridge (point-to-point link, e.g. 5 GHz directional antennas): it connects the two building switches transparently at Layer 2.

KEY POINTS **Exam tip: one role + one example per device**

GCE questions very often ask: "State the role of device X in this diagram." For each device, memorise one precise role and one example of use:

- NIC — connects a computer to the network and holds the MAC address (e.g. laptop Wi-Fi card).
- Switch — forwards frames to the destination port only using MAC addresses (e.g. school lab centre).
- Router — interconnects different networks, default gateway to the Internet, performs NAT (e.g. home gateway).
- Modem / ONT — converts signals for the ISP line (ADSL, cable, fibre) (e.g. CAMTEL ONT).
- Access Point — bridges Wi-Fi clients to the wired LAN (e.g. classroom ceiling AP).
- Hub — obsolete Layer 1 broadcaster (contrast with switch).
- Repeater/Extender — extends Wi-Fi coverage (half-duplex relay).
- Bridge — connects two LAN segments at Layer 2 (e.g. building-to-building wireless link).
- Patch panel / Keystone — structured cabling termination points.

- Crimping tool / Punch-down tool / Cable tester — installation and verification tools.

Never write that a hub and a switch do the same job.

Now that we know the equipment, the next section will put it into practice: physically setting up a computer network, from cabling the RJ45 connectors to configuring the machines.

4 SETTING UP A COMPUTER NETWORK

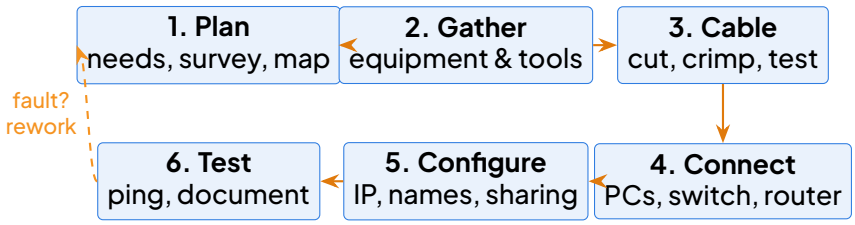
In the previous section we studied the **network equipment** — cables, connectors, switches, routers, network interface cards — and how to choose them. Owning good equipment, however, is not the same as having a working network. A box of switches and cables in the store of a school in Bamenda connects nobody. This section is the practical heart of the chapter: we learn, step by step, how to move from an empty computer room to a fully working **local area network** (LAN) — planning, cabling, connecting, configuring, testing and documenting. This is exactly the kind of task an ICT technician performs when a school, a cybercafé or an office in Douala asks for a network to be installed.

4.1 Overview of the Setup Process

Setting up a network is a project, and like every project it follows an ordered process. Skipping a step almost always means coming back later to fix avoidable mistakes.

METHOD **The 6-step procedure for setting up a small LAN**

1. Plan: assess needs, survey the site, draw the network plan.
2. Gather equipment: buy or collect cables, connectors, switch, router, tools, according to the plan.
3. Cable and crimp: measure, cut and terminate the cables with RJ45 plugs; test each cable.
4. Connect: plug computers into the switch, the switch into the router; label everything.
5. Configure: assign IP addresses, computer names, workgroup; enable sharing.
6. Test and document: verify connectivity (link lights, ping), fix faults, write the installation report.



The network setup process. If testing fails, return to the step where the fault originated.

4.2 Phase 1: Planning the Network

DEFINITION **Needs assessment**

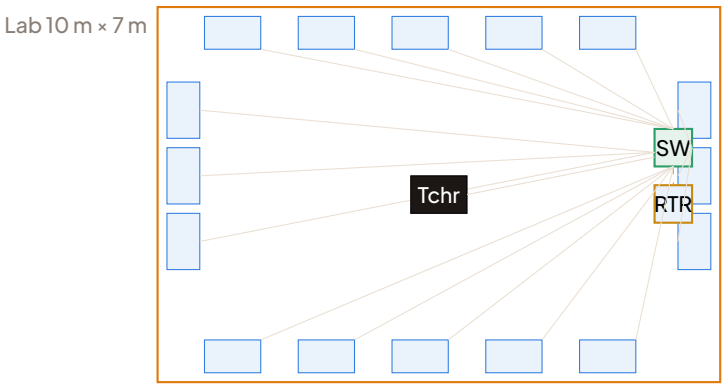
The **needs assessment** is the first stage of planning, in which we determine what the network must do: the number of users and devices, the services required (file sharing, printing, Internet access), the expected growth, and the available budget.

A good technician asks questions before buying anything. For a school computer laboratory the typical answers are: 15 student computers plus 1 teacher computer; shared printer; Internet access through a CAMTEL or MTN router; budget limited; possible extension to 30 computers next year.

The second planning activity is the **site survey**: visiting the room, measuring distances, locating electrical sockets, noting where cables can run (trunking along walls, ceiling, under desks) and choosing a central, dry, ventilated position for the switch. From the survey we draw the **network plan** — a scaled sketch showing each computer, the switch, the router and the cable runs. This plan is our working document and later becomes part of the documentation.

EXEMPLE **Planning a 15-computer school laboratory**

Room: 10 m×7 m. Computers arranged in three rows of five along the walls. Switch placed on a wall shelf at the centre of the room to minimise cable lengths (star topology). Router near the switch, connected to the ISP line. Longest cable run estimated at 12 m; we add 2 m slack per cable, so we prepare cables of about 14 m maximum.



Floor-plan sketch of the 15-computer laboratory: star topology, switch (SW) and router (RTR) on a central wall shelf, cable runs in trunking along the walls.

4.3 Phase 2: Choosing Topology and Equipment

From the previous sections, the **star topology** is the natural choice for a wired lab: easy to extend, a single cable failure affects only one machine, and switches are affordable. The equipment list follows directly from the plan.

Item	Qty	Specification / remark
Desktop computers with NIC	16	15 students + 1 teacher
Switch	1	24 ports (16 used, room to grow)
Router / modem	1	Provided by ISP (CAMTEL, MTN…)
UTP cable Cat 5e/6	1 box	305 m roll, solid or stranded
RJ45 connectors	50	More than 32 needed (spares for bad crimps)
Crimping tool, stripper	1 set	For RJ45
Cable tester	1	Checks continuity and wire order
Cable ties, trunking, labels	—	Cable management and identification

4.4 Phase 3: Cabling and Crimping

A UTP cable contains four twisted pairs, i.e. eight coloured wires. To connect it to a plug or socket, the wires must be inserted in a precise, standardised order. The standard used almost everywhere today is **T568B**.

METHOD **Crimping an RJ45 plug (T568B)**

1. Measure the cable run, add about 1 m of slack, and cut the cable cleanly.
2. Strip about 2 cm of the outer jacket with the stripper, without nicking the inner wires.

3. Untwist the pairs and arrange the eight wires flat, in the T568B order (pin 1 to pin 8): white-orange, orange, white-green, blue, white-blue, green, white-brown, brown.
4. Trim the wires straight so about 1.2 cm remains; keep the order tight.
5. Push the wires fully into the RJ45 plug (clip facing down, cable entering away from you); check through the plug that every wire reaches the end and the jacket enters the plug.
6. Insert the plug into the crimping tool and squeeze firmly once. Repeat for the other end, then test the cable with the cable tester: all eight lights must light in sequence on both ends.

WARNING **Straight-through vs crossover cable**

A **straight-through** cable (both ends T568B) connects unlike devices: computer to switch, switch to router. A **crossover** cable (one end T568A, the other T568B) connects like devices: computer to computer, or switch to switch without an uplink port. Many modern devices auto-sense (Auto-MDIX) and accept either cable, but in the GCE examination you must state the rule correctly: unlike devices → straight-through; like devices → crossover.

WARNING **Electrical safety**

Never crimp, punch down or dress cables on equipment that is powered on. Switch off and unplug devices before working inside them, keep cables away from mains power lines to avoid interference and shock risks, and never work with wet hands. A crimping tool and a cable knife are sharp: cut away from your body.

4.5 Phase 4: Physical Installation

With tested cables ready, installation is systematic:

1. Mount the switch on its shelf and the router beside it; connect their power adapters but keep them off until cabling is done.
2. Run each cable through the trunking from a computer's NIC to the switch; plug it into a switch port.
3. **Label both ends of every cable** and the corresponding switch port (e.g. PC01 → port 1). Labelling saves hours during troubleshooting.
4. Connect the switch to the router's LAN port with a straight-through cable; connect the router to the ISP line (telephone line, fibre box or 4G modem).

5. Power on the router, then the switch, then the computers. Check that the **link light** on each switch port and each NIC is on (usually green) or blinking: this confirms the physical connection.

4.6 Phase 5: Basic Configuration

Physical connection is not enough: each machine needs a logical identity on the network — an **IP address**. There are two ways to obtain one.

DEFINITION **DHCP**

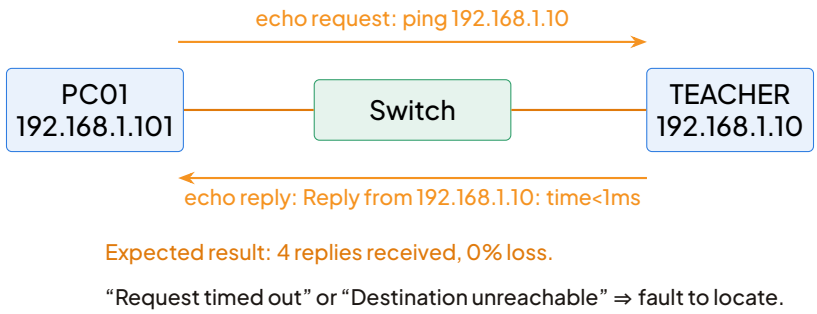
DHCP (Dynamic Host Configuration Protocol) is a network service that automatically assigns IP addresses (together with the subnet mask, default gateway and DNS server) to devices when they join a network. The alternative is **static addressing**, where the administrator types the address manually on each machine.

For our lab, the router usually acts as the DHCP server, so the simplest configuration is to set every computer to "Obtain an IP address automatically". Static addresses are preferred for machines that must always be found at the same address (the teacher's PC sharing the printer, a server). A typical private addressing scheme is 192.168.1.*x* with mask 255.255.255.0: the router takes 192.168.1.1, the teacher's PC 192.168.1.10 (static), and the students receive 192.168.1.100 and above from DHCP.

Each computer also receives a unique **computer name** (PC01, PC02, …, TEACHER) and joins the same **workgroup** (e.g. LABGROUP) so that machines can see one another. Finally, on the teacher's machine we enable **file and printer sharing** and share the printer and a folder "Course_Document so students can retrieve handouts and print.

4.7 Phase 6: Testing and Troubleshooting

The reference tool for testing connectivity is the **ping** command: it sends small echo-request packets to a target address and waits for replies. Open a command prompt and type, for example, `ping 192.168.1.10` from PC01.



Ping test between PC01 and the teacher's computer through the switch, with the expected replies.

METHOD **Locating a fault systematically**

1. Check the link lights on the NIC and the switch port. No light ⇒ physical problem (cable, crimping, port).
2. Swap the cable with a known-good one, or re-test it with the cable tester. A bad crimp (wire in wrong order, wire not reaching the end) is the most common fault: cut the plug and crimp again.
3. Check the cable type: straight-through for PC-to-switch.
4. `ping 127.0.0.1` tests the machine's own TCP/IP stack; then ping the gateway, then another PC.
5. Check the IP configuration: two machines with the same static IP cause an **IP conflict** — both may lose connectivity. Give a different address or use DHCP.

WORKED EXAMPLE 4 **Diagnosing a dead connection in the lab**

After installation, PC07 cannot reach the network. Its switch port link light is off, while all other ports are lit. The cable tester shows that wires 3 and 6 light up swapped. What is the fault and the remedy?

SOLUTION

The link light being off indicates a physical fault, not a configuration problem. The tester shows wires 3 and 6 (white-green and orange in a wrong crimp) are exchanged at one end: the RJ45 plug was crimped with two wires in the wrong order, so the pairs are broken and no link can be established. Remedy: cut off the faulty plug, re-arrange the eight wires in the correct T568B order (white-orange, orange, white-green, blue, white-blue, green, white-brown, brown), crimp a new plug, re-test with the cable tester until all eight wires light in sequence, then reconnect and verify with `ping 192.168.1.1`.

4.8 Safety, Cable Management and Documentation

A professional installation is recognised by its finishing. Cables are grouped with **cable ties** and hidden in **trunking**; nothing crosses walkways where people can trip; network cables are kept away from mains cables; the switch and router sit in a ventilated, dust-free position, ideally protected by a surge protector or UPS against the power cuts and surges common in our environment. Finally, the technician writes the **documentation**: the network plan, the list of equipment, the IP addressing table, the labelling scheme and the test results. This document is indispensable for maintenance and for any future extension of the network.

KEY POINTS **Key points**

- Setup follows six steps: plan, gather, cable, connect, configure, test.
- T568B wire order: white-orange, orange, white-green, blue, white-blue, green, white-brown, brown.
- Straight-through cable for unlike devices; crossover for like devices.
- DHCP assigns addresses automatically; static addresses for fixed machines.
- Link lights and `ping` are the first troubleshooting tools; bad crimping and IP conflicts are the classic faults.
- Label everything, manage cables safely, and document the installation.

EXERCISE 4 **Setting up a cybercafé network**

A cybercafé in Buea has 8 customer computers, 1 cashier computer, one shared printer and a 4G router. (a) Draw the network plan and state the topology. (b) Give the equipment list. (c) Propose an IP addressing scheme using DHCP with a static address for the cashier's PC. (d) After setup, PC05 replies to `ping 127.0.0.1` but not to `ping 192.168.1.1`. Suggest two possible causes and their remedies.

SOLUTION **Exercise 1**

(a) Star topology: all 9 computers connect to a central switch; the switch connects to the 4G router; the printer is shared from the cashier's PC (or a network printer on the switch). (b) Equipment: 9 computers with NICs, one 16-port switch, one 4G router, UTP Cat 5e/6 cable, at least 20 RJ45 plugs, crimping tool, cable tester, trunking, labels. (c) Network 192.168.1.0/24: router 192.168.1.1 as DHCP server (range 192.168.1.100–192.168.1.150), cashier PC static 192.168.1.10, mask 255.255.255.0, gateway 192.168.1.1. (d) Since the loopback ping works, TCP/IP is fine. Possible causes: (i) faulty cable or bad crimp between PC05 and the switch — test the cable and re-crimp; (ii) wrong IP configuration (wrong gateway or IP conflict) — set the PC to obtain an address automatically via DHCP, or correct the static settings, then ping the gateway again.

5 SELECTING EQUIPMENT FOR ORGANISATIONAL NETWORKS

In the previous section we learnt how to physically set up a small computer network: cabling, connecting devices, configuring IP addresses and testing connectivity. But before any cable is pulled, a harder question must be answered: which equipment should we buy, and how much of it? A school bursar, a cybercafé owner in Bamenda or a hospital administrator in Yaoundé does not buy a switch "because it looks nice". Equipment is selected to match **needs**, **growth** and **budget**. This section teaches you the professional reasoning behind equipment selection, ending with two complete case studies and an integration activity in which you produce a full **network dossier** — exactly the kind of scenario question the GCE Advanced Level examiners love.

5.1 Needs Analysis: The Starting Point

Imagine a tailor sewing a suit without measuring the customer. The result may be beautiful, but it will not fit. Selecting network equipment without analysing needs is the same mistake. A **needs analysis** is a systematic study of what the organisation actually requires from its network.

DEFINITION **Needs analysis**

A **needs analysis** is the systematic study, carried out before any purchase, of the users, services, traffic, growth prospects and physical constraints of an organisation, in order to determine the network equipment that must be acquired.

METHOD **Carrying out a needs analysis**

1. Count and classify the users. How many people will use the network? Are they students, secretaries, administrators, customers? Each class of user has different demands.
2. List the services required. File sharing, Internet access, shared printing, video streaming, VoIP telephony, a local server (e.g. school management software), video surveillance.
3. Estimate the traffic. Ten users typing documents generate little traffic; twenty-five users streaming video generate a lot. Traffic determines the required **bandwidth**.
4. Forecast growth. Will the organisation add users, computers or services in the next 3 to 5 years? A college that plans to open a computer science laboratory next year must plan for it today.
5. Note the physical constraints. Number of rooms and floors, distances between build-

ings, availability of electrical power, security of the equipment room.

EXEMPLE **Needs analysis for a small secondary school**

A school in Buea has: 1 administrative office (4 computers, 1 printer), 1 staff room (6 laptops), and 1 computer laboratory (30 student PCs + 1 teacher PC). Services needed: Internet sharing, file/print sharing, and a school-management database. Growth: a second laboratory is planned within 3 years. Conclusion: the network must serve about 42 stations today, scale to about 75, and carry moderate traffic (web + database, little video).

WARNING **Common mistake**

Analysing only today's situation. Equipment bought for exactly today's needs becomes undersized within two years, forcing a costly complete replacement. Always include a growth forecast in the needs analysis.

5.2 Sizing the Equipment

Once needs are known, each piece of equipment is sized — its capacity is chosen to match the demand.

Switches: port count and speed.

The two key figures are the **number of ports** and the **speed per port** (10/100/1000 Mb/s, i.e. Fast Ethernet or Gigabit Ethernet). Every wired station, printer, server and access point consumes one port; inter-switch links (uplinks) consume more.

PROPERTY **The 20% spare-port rule**

A switch (or a set of switches) should provide at least 20% more ports than the number of devices connected today, to allow future growth without replacing equipment. (This rule is a design guideline, not a theorem to prove; you must be able to apply it in calculations.)

EXEMPLE **Applying the 20% rule**

A laboratory has 30 student PCs, 1 teacher PC, 1 printer and 1 server: 33 devices, plus 1 uplink port to the rest of the network = 34 ports used. Adding 20%: $34 \times 1.2 = 40.8$, so at least 41 ports are required. Two 24-port switches (48 ports) satisfy the rule; a single

48-port switch also works but offers no redundancy if it fails.

Router capacity.

The router must handle the organisation's Internet traffic and the number of simultaneous users. For a small office, a combined router/modem from the ISP (CAMTEL, MTN, Orange) may suffice; a campus needs a router with higher throughput, multiple interfaces and possibly firewall and VPN features.

Wireless coverage.

Wireless access points (APs) have a limited indoor range (typically 20–50 m through walls). Sizing wireless means counting how many APs are needed to cover all rooms, and checking that each AP can serve the expected number of simultaneous clients.

WARNING **Common mistake**

Buying exactly the number of switch ports needed today. With zero spare ports, adding a single new computer forces the purchase of a whole new switch, plus re-cabling. The 20% rule exists precisely to avoid this trap.

5.3 Budgeting and Total Cost of Ownership

A network project has four cost families: **cabling** (cables, conduits, sockets, patch panels), **devices** (switches, router, APs, server, UPS), **labour** (technicians, configuration), and **operation** (electricity, ISP subscription, maintenance, replacements). The sum of purchase cost and operating cost over the equipment's lifetime is called the **total cost of ownership** (TCO). A cheap switch that fails every year may cost more in TCO than a reliable branded one. The art of budgeting is balancing performance against cost: Gigabit everywhere is wasteful for a typing pool, but Fast Ethernet is a bottleneck for a video-editing room.

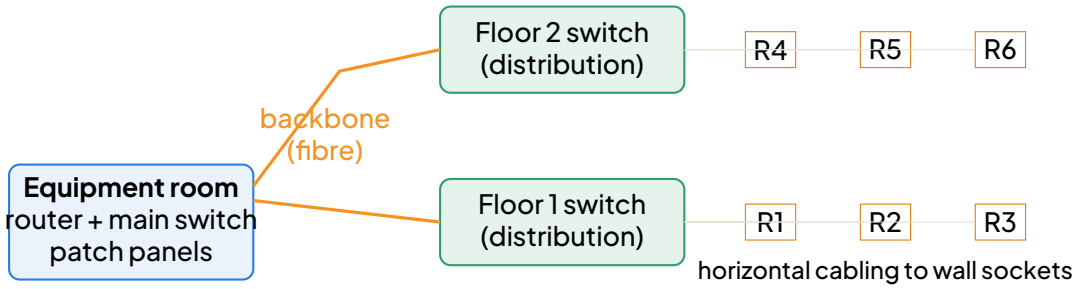
5.4 Structured Cabling for Multi-Room and Multi-Floor Organisations

When a network spans several rooms or floors, cables cannot be run randomly from desk to desk. The professional solution is **structured cabling**.

DEFINITION **Structured cabling**

Structured cabling is a standardised way of organising a building's network cabling into two subsystems: the **backbone** (vertical cabling linking equipment rooms and floors, often optical fibre) and the **horizontal cabling** (links from each floor's distribution switch to the wall sockets of individual rooms). Connections are made through **patch panels** in an **equipment room**, so any socket can be connected to any switch port simply by moving a patch cord.

The advantages are enormous: faults are located quickly, moving a user to another office requires no new cable, and the backbone can be upgraded (e.g. to faster fibre) without touching the horizontal cabling.



Structured cabling: equipment room, fibre backbone, floor distribution switches and horizontal cabling to wall sockets.

5.5 Case Study 1: A 25-Machine Cybercafé in Bamenda

EXPERIMENT **Equipping a cybercafé**

Needs analysis. 25 customer PCs + 1 cashier PC + 1 shared printer + 1 server (billing software) = 27 wired devices. Services: Internet browsing, printing, light video. Growth: up to 30 PCs within 2 years.
Topology. A single room ⇒ **star topology** around central switches (simple, cheap, one failure affects one machine).
Sizing. Ports needed today: 27 devices + 1 uplink between switches + 1 port to the router = 29. With 20% spare: $29 \times 1.2 = 34.8$, so at least 35 ports ⇒ two 24-port Gigabit switches (48 ports). One router with firewall; one wireless AP for customers' phones; a UPS for the server and cashier.

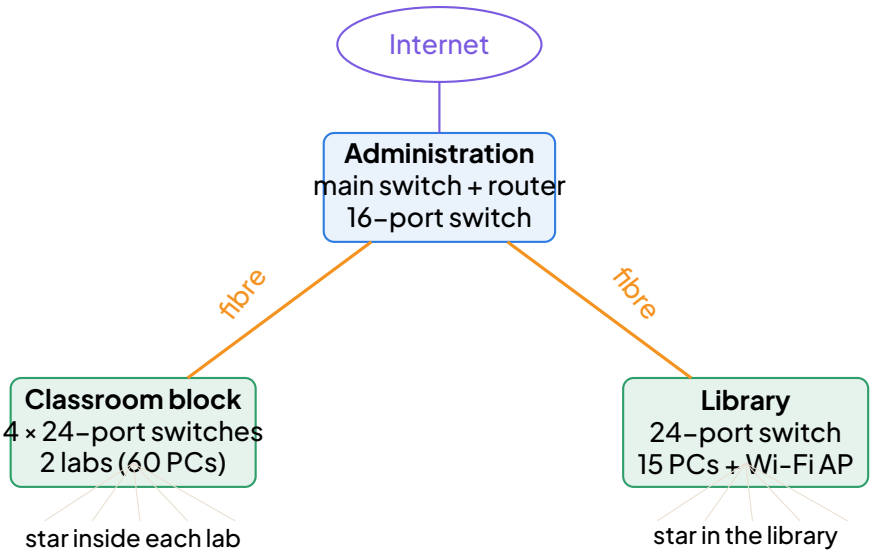
Item	Qty	Justification
24-port Gigabit switch	2	48 ports ≥ 35 required (20% rule)
Router with firewall	1	Shares the ISP link, protects the LAN
Wireless access point	1	Wi-Fi for customers' phones
Server + billing software	1	Controls sessions and payments
Cat6 cable (boxes) + sockets	3	Star cabling to 27 points
UPS	2	Protects server/cashier from power cuts
Labour (cabling + configuration)	—	Certified technician, testing

The estimated budget is then built by assigning a unit price to each line (obtained from local suppliers, e.g. in Bamenda's Commercial Avenue) and adding 10% contingency for unforeseen expenses.

5.6 Case Study 2: Interconnecting Three Buildings of a College Campus

EXPERIMENT **Campus network with fibre backbone**

Situation. A college has three buildings 80–150 m apart: Administration (10 stations), Classroom block (2 laboratories of 30 PCs), Library (15 stations + Wi-Fi).
Choice of topology. Copper Ethernet is limited to about 100 m per segment, so inter-building links use **optical fibre**. A main switch in the Administration equipment room connects to a distribution switch in each building: physically a **tree (hierarchical) topology**, each building internally wired as a star.
Sizing. Administration: 10 + 1 uplink = 11 ports used; 11 × 1.2 = 13.2, so at least 14 ports ⇒ one 16-port switch. Classroom block: 60 + 2 uplinks = 62 ports used; 62 × 1.2 = 74.4, so at least 75 ports ⇒ four 24-port switches (96 ports). Library: 15 + 1 AP +1 uplink = 17 ports used; 17 × 1.2 = 20.4, so at least 21 ports ⇒ one 24-port switch.



Three-building campus: tree topology with an optical-fibre backbone and a star inside each building.

5.7 The 5-Step Equipment Selection Procedure

METHOD **Selecting equipment for any organisation**

1. Analyse needs: users, services, traffic, growth, building layout.
2. Choose the topology: star for one room, tree with fibre backbone for several buildings.

3. Size the equipment: switch ports (apply the 20% rule) and speed, router capacity, wireless coverage.
4. Budget: cables, devices, labour, operation; compare performance vs cost using TCO.
5. Justify every choice: each line of the equipment list must be defended by a figure from steps 1–3.

WORKED EXAMPLE 5 **Sizing the switches of an accounting firm**

An accounting firm in Douala occupies one floor with 18 workstations, 2 network printers and 1 server, all wired, plus 1 wireless access point. The firm expects to hire 6 more staff within 3 years. Determine the number and size of switches required, justifying your answer.

SOLUTION

Step 1 — count ports used today: 18 + 2 + 1 + 1 = 22 devices, plus 1 uplink port to the router = 23 ports.
Step 2 — apply the 20% rule: $23 \times 1.2 = 27.6$, so at least 28 ports are required. The planned growth (6 staff) is covered: $23 + 6 = 29 \leq 48$.
Step 3 — choose: one 24-port switch is insufficient ($24 < 28$). Two 24-port Gigabit switches give 48 ports, satisfying the rule and the growth forecast, and providing partial redundancy. Gigabit is justified because accountants exchange large files with the server.
Conclusion: 2 switches of 24 Gigabit ports each.

EXERCISE 5 **A hospital network**

A district hospital has: Reception (4 PCs, 1 printer), Consultation wing (8 PCs), Laboratory (6 PCs, 2 printers), all in one building, plus a pharmacy in a separate block 120 m away (3 PCs, 1 printer). Internet access and a patient-record server are required. (a) Perform the needs analysis. (b) Choose a topology and cabling for the inter-building link, justifying your choice. (c) Size the switches using the 20% rule. (d) Sketch the network diagram.

SOLUTION **Exercise: a hospital network**

(a) Main building: 5 + 8 + 8 = 21 devices; pharmacy: 4 devices; plus 1 server and 1 AP ≈ 27 devices in total. Services: Internet, shared patient records, printing; growth moderate;

two buildings 120 m apart.
(b) Tree topology: a star in each block, optical fibre between them (copper Ethernet is limited to about 100 m per segment; 120 m exceeds it, and fibre is immune to electrical interference from medical equipment).
(c) Main building: 21 devices +1 server +1 AP +1 uplink = 24 ports used; 24 × 1.2 = 28.8, so at least 29 ports ⇒ two 16-port switches (32 ports) or one 24-port + one 8-port switch. Pharmacy: 4 + 1 uplink = 5 ports used; 5 × 1.2 = 6, so at least 6 ports ⇒ one 8-port switch.
(d) Diagram: router + main switch at reception, fibre link to the pharmacy's 8-port switch, stars to each room's wall sockets.

5.8 Integration: Producing a Complete Network Dossier

The final deliverable of any network project — and of any GCE scenario question — is a **network dossier** containing four elements:

1. **A labelled diagram** showing the topology, buildings, switches, router, servers and cabling types.
2. **An equipment list in a table**, each line justified by a calculation (port count, distance, bandwidth).
3. **A setup plan**: order of work (cabling → equipment room → device configuration → IP addressing → tests).
4. **A test checklist**: link lights on every port, `ping` between stations and to the router, Internet access, file and printer sharing, wireless coverage in every room, and a failover check (disconnect one cable and confirm only one station is affected).

KEY POINTS **Exam tip**

In scenario questions, always structure your answer as: diagram + equipment table + justification of each choice. Examiners award marks for the reasoning (needs analysis, 20% rule, distance limits), not for brand names or prices.

KEY POINTS **Key points of the section**

- Equipment selection starts from a **needs analysis**: users, services, traffic, growth, building layout.
- Size switches by port count (with at least 20% spare) and speed; size routers by throughput; size wireless by coverage.
- Budget with **total cost of ownership**, balancing performance against cost.

- Multi-room organisations use **structured cabling**: backbone + horizontal cabling + patch panels + equipment room.
- Inter-building links beyond about 100 m require **optical fibre** and a **tree topology**.
- The professional deliverable is a complete **network dossier**: diagram, justified equipment list, setup plan and test checklist.

6 INTEGRATION ACTIVITY

7 INTEGRATION ACTIVITY: DESIGNING AND IMPLEMENTING A SCHOOL NETWORK

7.1 Aims of the Activity

This integration activity mobilises, in one realistic project, all the competences developed in this chapter:

- Identify and compare **physical topologies** (bus, star, ring, mesh, tree) and justify the choice of a topology for a given organisation;
- Draw a clear, labelled **network diagram** from a site plan;
- Identify and select **network equipment** (switches, router, modem, access point, cables, connectors) according to needs and budget;
- Prepare and test a real **straight-through RJ45 cable** (crimping, wire order T568B, continuity test);
- Configure **IP addresses** on hosts and verify connectivity with `ping`;
- Work in a team, manage a budget in FCFA, and defend a technical dossier orally before a decision-making audience.

KEY POINTS **Success Criteria**

Your dossier is successful if: the topology is justified (not just named), the diagram is complete and labelled, every equipment item is quantified and priced, the total stays within the budget, at least one crimped cable passes the tester, and two configured PCs successfully ping each other.

DEFINITION **Physical Network Topology**

The **physical topology** of a network defines the geometric arrangement of the cables, devices and interconnections that make up the local area network (LAN). The five classic topologies are bus, star, ring, mesh and tree (hierarchical star). Each has distinct cost, reliability, scalability and maintenance characteristics.

PROPERTY **Comparison Criteria for Topology Selection**

When choosing a topology for an organisation, evaluate:

1. Cost: cable length, number of ports, active equipment.
2. Reliability / Fault tolerance: impact of a single cable or port failure.
3. Ease of maintenance / troubleshooting: isolation of faults, simplicity of moves/adds/changes.
4. Scalability: ease of adding new hosts or segments.
5. Performance: collision domains, bandwidth sharing, latency.

7.2 Situation: Networking Garrisson Secondary School

Garrisson Secondary School is a growing Anglophone secondary school in the North West Region of Cameroon. The school has two buildings separated by a courtyard:

- **Building A** contains Computer Lab 1 (20 PCs), Computer Lab 2 (20 PCs) and the Library (3 PCs used for cataloguing and research).
- **Building B**, located 60 m away from Building A, contains the Administration Office (5 PCs: Principal, Vice-Principal, Bursar and two Secretaries).

The Principal has just obtained an Internet subscription from a local ISP (a CAMTEL fibre offer, or a 4G router from MTN/Orange where fibre is unavailable). She wants:

1. All 48 computers to share files, a central printer in the Administration Office and the single Internet connection;
2. A Wi-Fi hotspot in the Library for students' laptops and phones;
3. A network that is easy to maintain and easy to extend (a third lab is planned next year);
4. The whole project to cost at most **1 500 000 FCFA**, including cabling and equipment.

The Principal has contacted your class. Working in teams of 3 to 5 students, you must produce a complete technical dossier, carry out a partial practical implementation in the school lab, and defend your proposal before the class, which plays the role of the school administration.

Data	Building A	Building B
Hosts	20 + 20 + 3 = 43 PCs	5 PCs + 1 shared printer
Distance to other building	60 m (aerial or buried cable across the courtyard)	
Special need	Wi-Fi hotspot in the Library	Shared printer for all users
Budget	1 500 000 FCFA maximum	

DID YOU KNOW? **Cameroon Context**

In Cameroon, most schools and SMEs rely on MTN or Orange for 4G/5G Internet backup, while CAMTEL provides fibre backbone in urban areas. The ANTIC (Agence Nationale des Technologies de l'Information et de la Communication) regulates cybersecurity and promotes ICT in education. A typical school LAN uses UTP Cat6 copper for indoor runs (≤100 m) and fibre for inter-building links to avoid lightning damage — a real concern during the rainy season in the North West Region.

7.3 Tasks

Task 1 — Site Plan and Topology Choice.

Draw a simple site plan of the two buildings. Choose a physical topology for each building and for the link between them. Justify your choice by comparing at least two topologies on the criteria of cost, reliability, ease of maintenance and scalability. Explain why a bus or a ring would (or would not) be appropriate here.

Task 2 — Labelled Network Diagram.

Produce the complete logical diagram of the network showing: the ISP modem, the router, the switches (with the number of ports used on each), the access point, the inter-building link and the hosts (you may represent groups of PCs symbolically, indicating the quantity). Label every device and every cable type.

Task 3 — Equipment List and Budget.

Establish a table of all equipment to buy: switches (state the port count of each and justify: e.g. a 24-port switch for a 20-PC lab leaves room for growth), router, modem, wireless access point, UTP cable (estimate the length: allow runs of 30 m average per PC inside a lab, plus the 60 m inter-building link — should you use UTP or fibre for this link? Justify), RJ45 connectors, patch panels or trunking, and a network printer if needed. Use realistic local prices (survey a shop or an online seller in your town) and check that the total does not exceed 1 500 000 FCFA.

Task 4 — Practical Cabling.

In the lab, crimp at least one straight-through RJ45 cable following the T568B wire order. Test it with a cable tester. Record the wire order you used and the test result in your dossier.

METHOD **Crimping a Straight-Through RJ45 Cable (T568B)**

1. Cut the UTP cable to the required length (+ 10 cm margin).
2. Strip 2.5 cm of the outer jacket without nicking the inner wires.
3. Untwist the four pairs and arrange them in the T568B order:

   1. White-Orange, 2. Orange, 3. White-Green, 4. Blue, 5. White-Blue, 6. Green, 7. White-Brown, 8. Brown.

4. Flatten and align the wires; trim them to 1.3 cm from the jacket.
5. Insert the wires fully into the RJ45 plug (tab facing down, pins 1–8 left to right).
6. Crimp firmly with a ratchet crimping tool.
7. Repeat on the other end (same T568B order for straight-through).
8. Test with a cable tester: all 8 LEDs must light in sequence on both master and remote units.

Task 5 — Configuration and Connectivity Test.

Propose an IP addressing plan for the school (network address, subnet mask, ranges per room, gateway address). Then, on two real PCs in the lab, configure static IP addresses from your plan and verify connectivity with `ping`. Note the commands used and the results.

METHOD **Static IP Configuration and Ping Test (Windows)**

1. Open `Control Panel >Network and Internet >Network and Sharing Center >Change adapter settings`.
2. Right-click the Ethernet adapter >`Properties` >select `Internet Protocol Version 4 (TCP/IPv4)` >`Properties`.
3. Choose `Use the following IP address` and enter:
   - IP address (e.g. `192.168.10.11`)
   - Subnet mask (`255.255.255.0`)
   - Default gateway (`192.168.10.1`)
   - Preferred DNS server (`8.8.8.8` or ISP DNS)
4. Click `OK` twice.
5. Open `Command Prompt` and run `ping <other_PC_IP>`. Verify replies with `TTL=128` (Windows) or `TTL=64` (Linux) and `time<1ms`.

Task 6 — Presentation and Defence.

Present your dossier in 10 minutes before the class (the “school administration”). Be ready to defend: your topology choice, your budget lines, and your plan for the future third lab.

WARNING **Common Mistakes to Avoid**

- Do not choose a topology "because it is the best" in general — justify it for this situation.
- Do not forget the inter-building link in the budget.
- Do not buy a 24-port switch for 24 hosts: keep spare ports for growth and uplink.
- Remember that UTP is limited to about 100 m per run (including patch cords).
- An IP plan where two rooms share the same host addresses, or where the gateway is missing, will fail the `ping` test.
- Do not mix T568A on one end and T568B on the other for a straight-through cable (that creates a crossover).

7.4 Model Answer

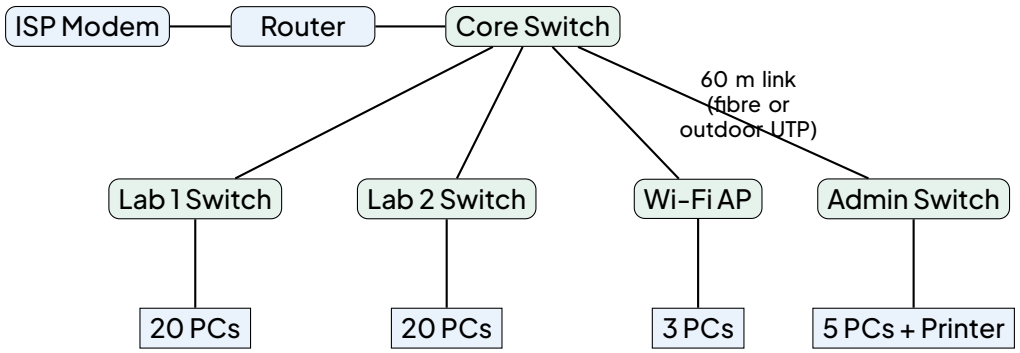
Task 1 — Topology Choice.

Inside each room we choose a **star topology**: every PC is connected by its own UTP cable to a central switch. Compared with a **bus** (one shared coaxial cable), the star costs slightly more in cable, but a single cable failure disconnects only one PC, faults are easy to localise, and adding a PC is trivial — essential since a third lab is planned. A **ring** would complicate maintenance and is not supported by standard Ethernet equipment (no commercial Ethernet ring switches for LANs). The two building switches are linked to a central (core) switch in Building A, giving an overall **tree (hierarchical star) topology**, which is the standard structure of organisational LANs.

EXEMPLE **Why Not Bus or Ring?**

Bus: A single break in the backbone cable brings down the entire segment. Troubleshooting requires checking the whole cable. Adding a host means tapping into the backbone (vampire taps or T-connectors) — impractical for UTP. Ring: Token Ring / FDDI are obsolete. Modern Ethernet does not use physical rings; a break in a ring also partitions the network unless a dual-ring (redundant) is used, which doubles cable cost.

Task 2 — Network Diagram.



Proposed tree topology for Garrisson Secondary School.

Task 3 — Equipment List and Budget (Indicative Prices from Bamenda/Buea Market Survey, 2024).

Item	Qty	Unit Price (FCFA)	Total (FCFA)
Switch 24 ports Gigabit (Labs 1 & 2)	2	45 000	90 000
Switch 8 ports Gigabit (Admin)	1	15 000	15 000
Core Switch 8 ports Gigabit (Building A)	1	15 000	15 000
Router (Gigabit WAN/LAN, VPN capable)	1	35 000	35 000
ISP Modem / 4G Router (MTN/Orange/CAMTEL)	1	25 000	25 000
Wireless Access Point (Ceiling mount, PoE)	1	30 000	30 000
UTP Cable Cat6, Box of 305 m (1000 ft)	3	40 000	120 000
Outdoor Fibre Link 60 m + 2 Media Converters	1	80 000	80 000
RJ45 Connectors (Bag of 100)	1	5 000	5 000
Trunking, Patch Cords, Crimping Tool, Tester	1	40 000	40 000
Network Printer (Admin, Wi-Fi + Ethernet)	1	150 000	150 000
Installation Labour & Contingency (10%)	1	100 000	100 000
Total			**705 000**

The total of about **705 000 FCFA** stays well within the 1 500 000 FCFA budget, leaving a comfortable margin for the future third lab (additional 24-port switch + cable box ≈ 85 000 FCFA). For the 60 m inter-building link, UTP Cat6 is technically acceptable (under the 100 m limit) if run in a protective conduit; however, **outdoor fibre with media converters** is strongly preferred for immunity to lightning, electromagnetic interference and ground loops — a critical consideration in the North West Region.

Task 4 — Cabling.

Straight-through cable, T568B order on both ends: White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown. After crimping, the tester lights wires 1 to 8 in sequence on both ends: the cable passes.

WORKED EXAMPLE 6 **Cable Test Interpretation**

Statement: A cable tester shows LEDs 1–8 lighting on the master unit, but on the remote unit LED 3 and LED 6 are swapped. What is the fault?

SOLUTION

Solution: The wire map shows that pins 3 and 6 are crossed. In T568B, pin 3 is White-Green and pin 6 is Green. This indicates that the Green pair (pins 3 & 6) has been reversed at one end. Re-crimp that end ensuring the correct order.

Task 5 — IP Plan and Test.

Private network `192.168.10.0/24` (mask `255.255.255.0`). Gateway (router LAN interface): `192.168.10.1`. Address allocation:

Area	IP Range	Notes
Lab 1 (20 PCs)	192.168.10.11 – 192.168.10.30	Static or DHCP pool
Lab 2 (20 PCs)	192.168.10.31 – 192.168.10.50	Static or DHCP pool
Library (3 PCs + Wi-Fi)	192.168.10.51 – 192.168.10.60	Wi-Fi clients via AP (DHCP)
Admin (5 PCs)	192.168.10.61 – 192.168.10.65	Static for servers/printers
Network Printer	192.168.10.70	Static
Reserved / Future	192.168.10.71 – 192.168.10.254	Growth

Test: PC1 set to `192.168.10.11`, PC2 to `192.168.10.31`. From PC1: `ping 192.168.10.31` replies successfully (4 packets sent, 4 received, 0% loss, time <1 ms), confirming Layer 3 connectivity across the switches and core.

Task 6 — Defence.

The team presents the diagram, budget and live ping demonstration, and explains that the tree-of-stars design lets the future third lab be added by simply connecting one more 24-port switch to an available port on the core switch, with no re-cabling of existing labs. The budget margin of ≈ 795 000 FCFA covers the third lab and unforeseen expenses.

EXERCISE 6 **Extension: VLAN Segmentation (Optional for Advanced Students)**

The school later decides to separate student traffic (labs, library) from admin traffic for security. Propose a VLAN plan (VLAN IDs, IP subnets, trunk ports) and explain which device must perform inter-VLAN routing.

SOLUTION **Extension**

- VLAN 10 (Students): 192.168.10.0/24 — ports on Lab 1, Lab 2, Library switches + AP.

- VLAN 20 (Admin): 192.168.20.0/24 — ports on Admin switch.
- Trunk links: Core–Lab1, Core–Lab2, Core–Admin, Core–AP (tagged).
- Inter-VLAN routing: Router (router-on-a-stick) or Layer 3 core switch. The router's LAN interface is configured with sub-interfaces for VLAN 10 and VLAN 20, each with its own gateway IP (e.g. 192.168.10.1 and 192.168.20.1).

8 METHODS AND WORKED EXAMPLES

9 PHYSICAL NETWORK TOPOLOGIES

DEFINITION **Physical Network Topology**

The **physical topology** of a network describes the actual geometric layout of cables, devices, and connection points. It differs from the **logical topology**, which describes how signals travel and how devices communicate (e.g., a physical star wired to a hub behaves logically as a shared bus).

PROPERTY **Six Core Physical Topologies**

The syllabus requires mastery of these six fundamental topologies:

- Bus (Linear Bus): All nodes connect via drop lines/taps to a single backbone cable (historically coaxial 10BASE2/10BASE5, later UTP in a daisy-chain). Requires terminators at both ends to absorb signals and prevent reflections.
- Star: Every node connects to a central device (hub, switch, or wireless access point) via a dedicated point-to-point link. The central device manages all traffic.
- Ring: Each node connects to exactly two neighbours, forming a closed loop. Data travels unidirectionally (single ring) or bidirectionally (dual ring for redundancy). Token passing or similar media access control is typical.
- Mesh: Nodes interconnect with multiple redundant paths. Full mesh: every node connects directly to every other node ($n(n-1)/2$ links). Partial mesh: only critical nodes have multiple interconnections.
- Tree (Hierarchical/Extended Star): Groups of star-configured networks connected to a linear bus backbone or to a higher-level central device. Forms a hierarchy: core → distribution → access.
- Hybrid: A combination of two or more basic topologies (e.g., star-bus, star-ring, star-mesh). Modern enterprise networks are almost always hybrid.

METHOD **Identifying and Describing Physical Network Topologies**

1. List the six core topologies: Bus, Star, Ring, Mesh, Tree (hierarchical), Hybrid.
2. Recognise each by its physical layout:
   - Bus: Single backbone cable (coaxial or twisted pair) with drop lines/taps to each node.
   - Star: Every node connects to a central device (hub/switch) via dedicated cable.
   - Ring: Each node connects to exactly two neighbours forming a closed loop; data travels in one direction (or both in dual-ring).
   - Mesh: Every node connects directly to every other node (full mesh) or to several others (partial mesh).
   - Tree: Groups of star-configured networks connected to a linear bus backbone (hierarchy).
   - Hybrid: Combination of two or more basic topologies (e.g., star-ring, star-bus).
3. Key parameters to note for each: Number of cables/ports, cable length, single point of failure, ease of adding/removing nodes, typical media (UTP, fibre, coaxial).
4. Reflex: When shown a diagram, trace the physical cables — not the logical signal path — to name the topology.

EXEMPLE **Topology Identification Practice**

Scenario: A network diagram shows 8 computers. Four computers on the left connect to Switch A. Four computers on the right connect to Switch B. Switch A and Switch B are linked by a single fibre cable. No other connections exist.

1. What is the physical topology?
2. How many cable links are needed for the 8 computers?
3. Identify all single points of failure.

Solution:

1. Hybrid (Star-Bus / Star-Fibre Backbone) — two stars interconnected by a backbone link.
2. Each computer → its switch: 8 UTP links. Inter-switch: 1 fibre link. Total = 9 distinct cable links.
3. Each switch is a local SPOF for its segment. The inter-switch fibre link is a global SPOF: if cut, the two stars cannot communicate (though each remains internally functional).

WORKED EXAMPLE 7 **Identifying Topologies from Diagrams and Descriptions**

Statement: Three physical network layouts are described below. For each, (i) name the topology, (ii) state the number of cable links required for 6 computers, (iii) identify the single point of failure (if any), and (iv) give one typical use case in a Cameroonian context.

1. All computers connect to a central switch using UTP patch cords. The switch is powered by mains electricity with no UPS.
2. Computers are daisy-chained along a single coaxial cable terminated at both ends with 50 Ω terminators. A T-connector attaches each NIC to the backbone.
3. Two separate star networks (each with its own switch) are interconnected by a single fibre link between the two switches. Each star serves a different department.

SOLUTION

Detailed Solution:
Layout 1:

1. Topology: **Star** (physical star, logical star).
2. Cable links for 6 computers: Each computer needs one patch cord to the switch → 6 links. The switch requires at least 6 ports (an 8-port switch is typical).
3. Single point of failure: The central switch (and its power supply). If the switch fails or loses power, the entire segment goes down. No UPS exacerbates this.
4. Cameroonian use case: A typical cyber café in Yaoundé or Douala with 10–20 PCs connected to a central switch; also a school computer lab (e.g., Lycée Bilingue d'Essos) where the teacher's station manages the class via the switch.

Layout 2:

1. Topology: **Bus** (linear bus, historically "Ethernet 10BASE2" or "Thinnet").
2. Cable links for 6 computers: One continuous backbone cable with 6 T-connectors → effectively 1 backbone segment + 6 drop cables (very short). In terms of distinct cable runs: 1 main run + 6 drops.
3. Single point of failure: A break anywhere on the backbone (or a missing/loose terminator) brings down the whole network. Also, a faulty NIC that jams the bus can disrupt all.
4. Cameroonian use case: Legacy installation in a small administrative office in Buea using old coaxial cabling; or a temporary exam registration centre where laptops are quickly daisy-chained with BNC T-connectors.

Layout 3:

1. Topology: **Hybrid** (specifically Star-Bus or Star-Fibre Backbone). Two stars linked by a high-speed backbone.

2. Cable links for 6 computers (3 per star): Each star: 3 UTP links to its switch. Backbone: 1 fibre link between switches. Total distinct links = 3 + 3 + 1 = 7.
3. Single point of failure: Each star has its own switch as a local SPOF. The inter-switch fibre link is a global SPOF: if cut, the two departments cannot communicate (though each star remains internally functional).
4. Cameroonian use case: University campus (e.g., University of Buea) where Faculty of Engineering (star) and Faculty of Science (star) are linked by fibre; or a bank with two branches in Douala (Akwa and Bonanjo) each wired as a star, connected via CAMTEL fibre lease line.

Examiner's Tips:

- Always distinguish physical topology (cable layout) from logical topology (signal flow). A physical star wired to a hub behaves logically as a bus (shared medium).
- For bus topology, remember the terminators — they absorb signals to prevent reflections. Missing terminator → network errors.
- Hybrid topologies are the norm in modern organisations; pure bus or ring are rare in new installations.

KEY POINTS **Key Points: Physical Topologies**

- Physical topology = cable layout; Logical topology = signal flow.
- Star is the dominant LAN topology today (switched Ethernet).
- Bus and Ring are legacy (10BASE2, Token Ring, FDDI) — know them for exams but do not recommend for new installs.
- Mesh provides highest redundancy but at highest cost (cables, ports).
- Tree/Hierarchical = standard for multi-floor, multi-building, or multi-department sites.
- Hybrid = reality. Always describe the components (e.g., "star of stars with fibre backbone").

10 COMPARING AND CHOOSING A TOPOLOGY

METHOD **Comparing and Choosing a Network Topology**

1. Define the decision criteria: Cost (cable, ports, devices), Scalability (ease of adding nodes), Fault Tolerance (impact of a link/node failure), Performance (bandwidth sharing, collisions), Installation/Maintenance complexity.
2. Create a comparison table with topologies as rows and criteria as columns. Fill with qualitative ratings (High/Medium/Low) or quantitative estimates (e.g., cable length $\approx n \times d$ for star vs. $\approx (n-1) \times d$ for bus).
3. Match scenario constraints to topology strengths:
   - Low budget, few nodes, temporary: Bus (if legacy) or Star with a cheap unmanaged switch.
   - High reliability required (bank, hospital): Mesh (core) or Star with redundant switches/links.
   - Multiple departments, hierarchical management: Tree / Extended Star.
   - Existing infrastructure (old building): Hybrid — reuse whatever cabling exists.
4. Justify your choice in writing: "I recommend Extended Star because…" citing at least two criteria from the table.
5. Reflex: Never recommend a pure bus or ring for a new installation in 2025 — they are obsolete for LANs. Always assume switched Ethernet (star) as the baseline.

WORKED EXAMPLE 8 **Choosing a Topology for a New Office in Bamenda**

Statement: A new NGO office in Bamenda needs a LAN for 25 workstations, 3 network printers, and 2 servers. The building has two floors. Requirements:

- Budget is tight (donor-funded), but reliability is important for donor reporting.
- The IT officer (only one) must be able to troubleshoot quickly.
- Future expansion to 40 workstations is expected within 2 years.
- The building has conduit trunking already installed on each floor.

1. Draw a comparison table (Star vs. Tree vs. Mesh) for the criteria: Cable Cost, Port Count, Fault Isolation, Ease of Expansion, Troubleshooting Simplicity.
2. Recommend the best topology and justify with three bullet points.
3. Sketch a labelled physical diagram (text description acceptable) showing switches, uplink, servers, and workstations.

SOLUTION

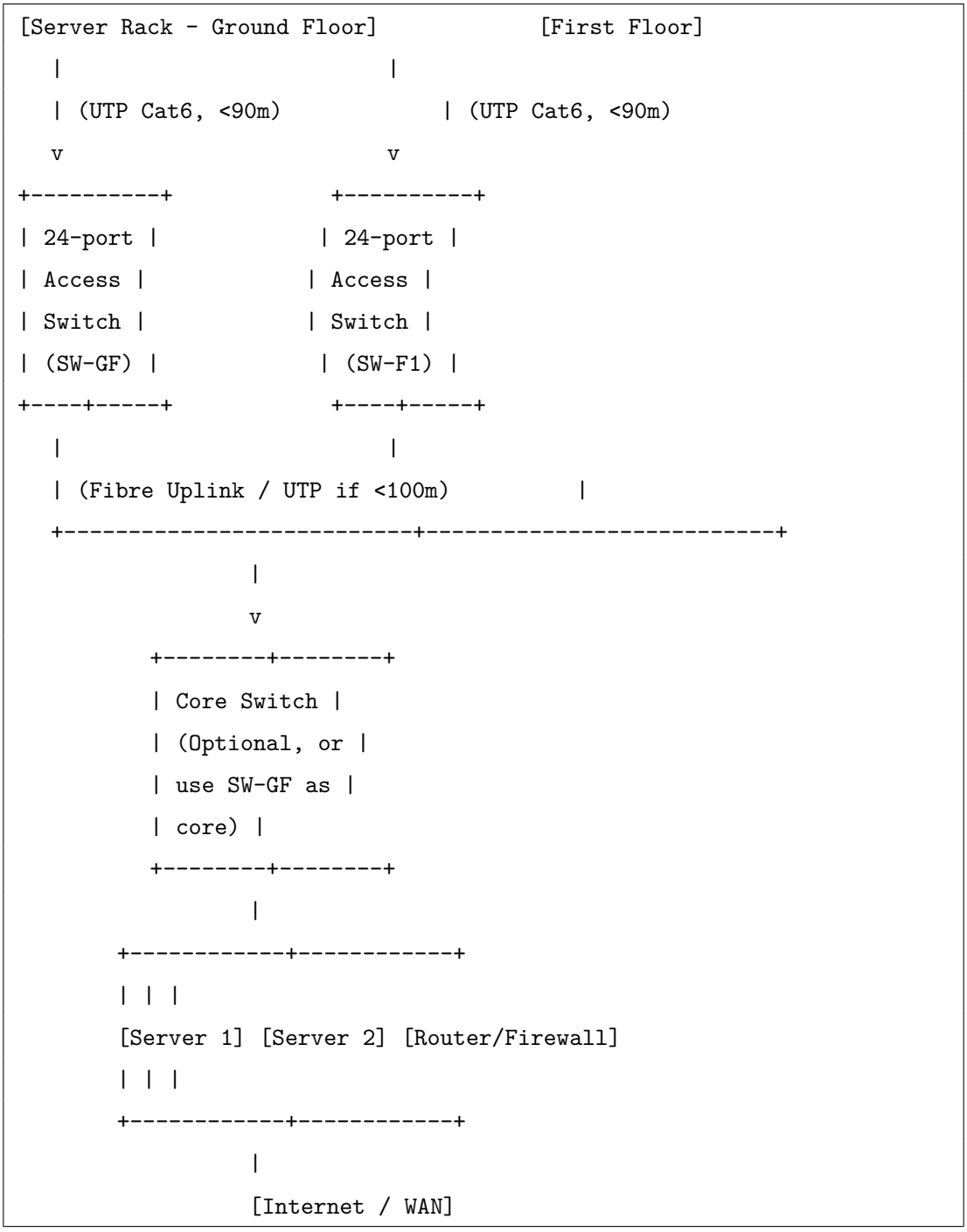
Detailed Solution:
1. Comparison Table

Criterion	Star (Single Switch)	Tree / Extended Star	Full Mesh
Cable Cost	Medium (each node to central switch)	Medium–High (backbone + horizontal)	Very High ($n(n-1)/2$ links)
Port Count	30+ ports on one switch (needs 48-port)	Two 24-port switches + uplink ports	29 ports per node (impossible)
Fault Isolation	Excellent (one cable = one node)	Excellent (same) + backbone risk	Excellent (multiple paths)
Ease of Expansion	Limited by switch ports; need new switch	Add switch on new floor/segment easily	Impractical (rewire all)
Troubleshooting	Simple: check link lights on one switch	Simple per segment; backbone separate	Complex: many interconnections

2. Recommendation: Tree / Extended Star (Two-floor hierarchical star).
Justification:

- Scalability: Each floor gets its own 24-port access switch (one per floor). Adding a third floor or more ports only requires adding another access switch and uplinking to a core switch — no rewiring of existing drops.
- Fault Isolation & Troubleshooting: A faulty cable or NIC affects only that workstation. The IT officer can see link lights on the floor switch immediately. The uplink fibre between floors is a single monitored link.
- Cost-Effectiveness: Uses two affordable 24-port unmanaged/managed switches (e.g., TP-Link TL-SG1024 or Cisco SG250) and one fibre uplink (or UTP if distance < 100 m). Full mesh is financially impossible; single 48-port switch would require long cable runs from both floors to a central cabinet, exceeding 100 m UTP limit for ground-floor nodes.

3. Physical Diagram Description (Text):



Notes:

- If budget is extremely tight, SW-GF can act as core (uplink from SW-F1 plugs into SW-GF). This saves one switch but makes SW-GF a SPOF for inter-floor traffic.
- Use Cat6 UTP for all horizontal runs (max 90 m in conduit + 10 m patch cords). Use multimode fibre (OM3/OM4) for inter-floor uplink if distance > 100 m or to avoid EMI in electrical trunking.

- Label every cable at both ends (e.g., “GF-01”, “F1-12”) and document in a spreadsheet — essential for the lone IT officer.

WARNING **Common Mistakes in Topology Selection**

- Recommending Bus or Ring for a new LAN — they are obsolete, lack fault isolation, and do not support modern switched Ethernet speeds.
- Ignoring the 100 m UTP distance limit — placing a single switch in a central cabinet for a multi-floor building often exceeds this limit for remote nodes.
- Confusing physical and logical topology — a star wired to a hub is physically a star but logically a shared bus (half-duplex, collisions).
- Forgetting growth — a 24-port switch filled today leaves no room for tomorrow’s hires.
- Over-engineering: proposing full mesh for a 20-node LAN — cost and complexity are unjustifiable.

11 NETWORK EQUIPMENT

DEFINITION **Network Equipment by OSI Layer**

Network devices operate at specific layers of the OSI model. Understanding the layer determines the device’s function, addressing scheme, and configuration complexity.

PROPERTY **Equipment Classification by OSI Layer**

- Layer 1 (Physical): Cables (UTP Cat5e/6/6a, Fibre OM3/OM4, Coaxial), Connectors (RJ-45, LC/SC/ST, BNC), Patch panels, Wall plates, Repeaters/Hubs (obsolete), Media converters.
- Layer 2 (Data Link): NIC (Network Interface Card), Switch (multi-port bridge), Wireless Access Point (WAP), Bridge. Forward frames based on MAC addresses.
- Layer 3 (Network): Router, Layer-3 Switch, Firewall (often L3/L4), Modem (ISP demarcation). Forward packets based on IP addresses; perform routing, NAT, ACL,

DHCP.

- Layer 4–7: Load balancer, Proxy, WAF, Application Delivery Controller. Inspect payload, session, or application data.

METHOD **Identifying Network Equipment and Their Roles**

1. Categorise equipment by OSI layer (see table above).
2. Key functional distinctions:
   - Hub vs. Switch: Hub = shared bandwidth, half-duplex, one collision domain. Switch = dedicated bandwidth per port, full-duplex, separate collision domains, learns MAC addresses.
   - Switch vs. Router: Switch forwards frames within same LAN (MAC). Router forwards packets between networks (IP), performs NAT, ACL, DHCP.
   - Modem vs. Router: Modem converts ISP signal (DSL, Cable, Fibre ONT, 4G/5G) to Ethernet. Router does IP routing. Many ISP boxes (MTN/Orange Flybox, CAMTEL GPON ONT+Router) combine both.
   - Access Point vs. Router: AP bridges wireless to wired (Layer 2). Wireless Router = Router + Switch + AP in one box.
3. Reflex: When asked "which device…", first decide the layer/function: connect PCs in one room → Switch; connect LAN to internet → Router; extend Wi-Fi → AP (not another router in NAT mode).

EXEMPLE **Device Selection Quick Reference**

Need	Device	Key Feature
Connect 20 PCs in a lab	24-port Gigabit Switch (L2)	MAC learning, full-duplex, VLAN support
Connect LAN to CAMTEL fibre	Router / Gateway	PPPoE client, NAT, Firewall, DHCP server
Provide Wi-Fi in a hall	Ceiling AP (PoE) + PoE Switch/Injector	802.11ax (Wi-Fi 6), VLAN tagging, roaming
Connect two buildings 500 m apart	Media Converters + Fibre (or Fibre SFP in switches)	OM3/OM4 multi-mode or OS2 single-mode
Load balance two ISP links	Dual-WAN Router / SD-WAN appliance	Failover, load balancing, policy routing

WORKED EXAMPLE 9 **Selecting the Right Device for Specific Functions**

Statement: A small hotel in Limbe wants to provide guest Wi-Fi and wired internet for the reception desk, manager's office, and conference room. They have a CAMTEL fibre subscription (GPON ONT provided). The manager buys the following equipment from a local vendor:

- Device A: 8-port Gigabit Unmanaged Switch (TP-Link TL-SG108)
- Device B: Wireless Router (TP-Link Archer C6) — labelled "Router/AP/Repeater"
- Device C: PoE Injector (for a single ceiling AP)
- Device D: USB 4G Dongle (Huawei E3372) — "for backup"

1. For each device, state its primary OSI layer(s) and its role in this network.
2. The manager plugs the ONT into Device B's WAN port, connects Device A to a LAN port of Device B, and plugs the reception PC, office PC, and conference room PC into Device A. He sets Device B to "Wireless Router" mode. Guests complain they can access the reception PC's shared folder. Explain why and how to fix it using the existing hardware.
3. The manager wants to add a ceiling-mounted AP in the garden for better coverage. He has a PoE-capable AP (Ubiquiti U6-Lite) and Device C (PoE Injector). Describe the correct cabling from Device A to the AP.
4. If the CAMTEL fibre fails, how can Device D be used for automatic failover? What additional feature must Device B support?

SOLUTION

Detailed Solution:
1. Device Roles and OSI Layers

Device	Primary OSI Layer(s)	Role in This Network
A: 8-port Switch	Layer 2 (Data Link)	Expands wired ports: connects reception, office, conference PCs to the router's LAN. Creates separate collision domains, learns MACs.
B: Wireless Router	Layer 3 (Network) + Layer 2 (Switch/AP)	Edge router: terminates CAMTEL ONT (WAN), does NAT, DHCP, firewall for entire hotel LAN. Built-in AP: provides guest Wi-Fi. Built-in 4-port switch: connects Device A and other wired devices.
C: PoE Injector	Layer 1 (Physical)	Injects DC power (48V) onto Ethernet cable to power the ceiling AP (Device C has no data processing role).
D: 4G USB Dongle	Layer 1 + Layer 2 (Modem)	Backup WAN link: provides cellular internet (MTN/Orange 4G) when fibre fails. Appears as Ethernet or RNDIS interface to router.

2. Guest Access to Reception PC — Cause and Fix

- Cause: Device B is in "Wireless Router" mode (default). Its built-in switch ports (LAN1–LAN4) and Wi-Fi interfaces are bridged together in the same LAN subnet (e.g., 192.168.0.0/24). Guests on Wi-Fi and wired PCs on Device A (connected to LAN port of B) are in the same broadcast domain. No firewall rule isolates them → guests can SMB-browse to reception PC.
- Fix using existing hardware: Enable Guest Network (or "AP Isolation" / "Client Isolation") on Device B's Wi-Fi settings. This creates a separate virtual SSID (e.g., "Hotel_Guest") with its own subnet (e.g., 192.168.1.0/24) and firewall rules blocking access to the main LAN. Do not plug the AP into a separate router (double NAT) — the

Archer C6 supports Guest Network natively.

- Alternative (if no Guest Network): Put Device B in Access Point mode, buy a separate router for the main LAN, and use VLANs — but that requires new hardware. Guest Network is the correct answer here.

3. Cabling for Ceiling AP with PoE Injector

1. Run a single Cat6 UTP cable (max 90 m solid + 5 m patch each end) from a free port on Device A (Switch) to the PoE Injector's "Data In" / "LAN" port (located indoors, near switch).
2. Connect a short patch cord from Injector's "PoE Out" / "Data & Power" port to the AP's Ethernet port (the cable continues through ceiling conduit to AP).
3. Critical: The injector must be 802.3af/at compliant (the U6-Lite needs 802.3af). The TP-Link injector (TL-POE150S or similar) works. Do not use a passive 24V injector — it will damage the Ubiquiti AP.
4. Power the injector from a mains socket (preferably on UPS).

4. 4G Failover with Device D

- Requirement: Device B (Archer C6) must support WAN Failover / Dual WAN / Backup WAN with a USB WAN or Ethernet WAN port configurable for failover.
- Setup: Plug Device D (4G dongle) into Device B's USB port. In Device B's web UI: Network → Internet → Backup WAN → select "USB Modem", configure APN (e.g., "mtncameroon.net" or "orange.cm"), enable "Failover Mode" (primary = WAN port / fibre, backup = USB). Set detection method (ping to 8.8.8.8 every 30 s).
- Note: The Archer C6 v3+ supports USB modem failover. If the model lacks this, a separate router with dual WAN (e.g., MikroTik hAP ac², Ubiquiti EdgeRouter X) is needed — Device D alone cannot trigger failover.

DID YOU KNOW? **Cameroon ISP Equipment Context**

- CAMTEL GPON: Provides an ONT (Optical Network Terminal) — typically Huawei or ZTE — with one or more Gigabit Ethernet ports. The ONT is Layer 1/2; you connect your router's WAN port to it. PPPoE credentials are required.
- MTN/Orange Fibre: Often provide a combined ONT+Router ("box"). You can request "bridge mode" to use your own router, or use DMZ/port forwarding.
- 4G/5G Backup: MTN and Orange sell USB dongles (Huawei, ZTE) and MiFi routers. For business, ask for a "Fixed Wireless Access" (FWA) router with external antenna ports — more stable than a dongle.
- ANTIC Compliance: All network equipment handling personal data must comply with

Law No. 2010/012 on Cybersecurity and Cybercrime. Logs must be kept for 12 months; incident reporting to ANTIC is mandatory.

12 SETTING UP A COMPUTER NETWORK

METHOD **Setting Up a Small Computer Network (Step-by-Step)**

1. Phase 1 — Planning: Draw topology, list devices, assign IP scheme (private range, subnet mask, gateway, DNS), label cables.
2. Phase 2 — Physical Installation:
   (a) Install switches/router in ventilated, secure cabinet/rack.
   (b) Run horizontal cables (Cat6) from wall plates to patch panel (or directly to switch). Respect 100 m max (90 m + 10 m patches).
   (c) Terminate cables: TIA/EIA-568B colour code on both ends (patch panel + RJ-45 plugs). Test every link with cable tester (wiremap, length, attenuation).
   (d) Connect patch cords: Patch panel → Switch; Wall plate → PC/Printer.
   (e) Connect WAN: ONT/Modem → Router WAN.
   (f) Power on: Modem → Router → Switches → APs.
3. Phase 3 — Logical Configuration:
   (a) Access router web UI (default IP, admin/admin). Change admin password immediately.
   (b) Configure WAN: PPPoE (CAMTEL), DHCP (MTN/Orange), or Static IP.
   (c) Configure LAN: Private subnet (e.g., 192.168.10.0/24), enable DHCP server (pool .100–.200), set DNS (1.1.1.1, 8.8.8.8).
   (d) Configure Wi-Fi: SSID, WPA3-Personal (or WPA2-AES), strong passphrase, hide SSID optional.
   (e) Configure Guest Network (isolated subnet, bandwidth limit).
   (f) Assign static IPs / DHCP reservations for servers, printers, APs.
4. Phase 4 — Testing & Documentation:
   (a) Ping gateway from each PC. Ping 8.8.8.8. Browse web.
   (b) Test printer, file share, speed test (iperf3 or fast.com).

(c) Document: IP plan, device passwords (in sealed envelope), cable map, Wi-Fi channels.

5. Reflex: Always test before closing trunking/ceilings. Label both ends of every cable. Use a UPS for router + core switch.

WORKED EXAMPLE 10 **Complete Setup of a 10-PC Office Network in Douala**

Statement: You are contracted to set up a LAN for a law firm in Douala (Akwa). Requirements:

- 10 workstations (Windows 11), 1 network printer (HP LaserJet), 1 file server (Ubuntu Server).
- Internet via MTN Fibre (GPON ONT provided, PPPoE credentials given).
- Wired connections for all fixed devices; Wi-Fi for lawyers' laptops and guests.
- Budget allows: 1× 16-port Gigabit Switch (managed), 1× Wireless Router (dual-band, VLAN support), 1× Ceiling AP (PoE), Cat6 cable, RJ-45, patch panel, 6U wall cabinet, UPS (1000 VA).

1. Propose an IP addressing plan (subnet, gateway, DHCP pool, static reservations, VLANs for Guest/Staff).
2. List the exact cabling steps from ONT to switch to devices, including patch panel termination standard.
3. Write the router configuration checklist (WAN, LAN, VLANs, DHCP, Firewall, Wi-Fi, Admin security).
4. Describe how you would test and document the installation for handover.

SOLUTION

Detailed Solution:
1. IP Addressing Plan (VLAN-Segmented)

VLAN / Network	Subnet / CIDR	Gateway	Usage / DHCP Pool
VLAN 10 — Staff Wired	192.168.10.0/24	192.168.10.1	Workstations, Printer, Server. DHCP: .100–.150. Reservations: Server .10, Printer .11.
VLAN 20 — Staff Wi-Fi	192.168.20.0/24	192.168.20.1	Lawyers' laptops, phones. DHCP: .100–.150. Same firewall rules as VLAN 10 (inter-VLAN routing allowed).
VLAN 30 — Guest Wi-Fi	192.168.30.0/24	192.168.30.1	Guests only. DHCP: .100–.200. No access to VLAN 10/20. Internet only. Bandwidth limit: 5 Mbps/user.
Management (untagged on switch)	192.168.99.0/24	192.168.99.1	Switch, Router, AP management IPs. No DHCP. Static only.

2. Cabling Steps (Physical Installation)

1. Cabinet mounting: Fix 6U cabinet on wall in secure room (server room / storage). Install patch panel (16-port), switch, router, UPS inside.
2. Horizontal runs: Pull Cat6 UTP from cabinet to each wall outlet (10 workstations + 1 printer + 1 server + 2 AP locations = 14 drops). Use conduit/trunking. Leave 1 m slack at each end.

3. Termination at patch panel (cabinet end): Strip jacket, untwist pairs <13 mm, terminate on 110-type IDC blocks following TIA/EIA-568B colour code:

Pin	Colour (568B)
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

Punch down with Krone tool. Label each port (WS-01 … WS-10, PRN-01, SRV-01, AP-01, AP-02).

4. Termination at wall plates (room end): Same 568B on keystone jacks. Snap into face-plates.
5. Patch cords (cabinet): Factory-made Cat6 patch cords (1 m) from patch panel ports → Switch ports. Match labels.
6. Device patch cords (rooms): 2–3 m Cat6 patch cords from wall plate → PC/Printer/Server.
7. AP cabling: Run one drop to ceiling AP location. At cabinet, patch panel port → PoE injector "Data In" → PoE injector "Data+Power Out" → patch cord to AP (or direct if switch is PoE+). Second AP (router's built-in) needs no extra cable.
8. WAN: ONT (GPON) → Router WAN port (Cat6 patch cord).
9. Power: Router, Switch, PoE Injector → UPS battery-backed outlets. ONT → UPS if possible.

3. Router Configuration Checklist (Managed Router, e.g., MikroTik hAP ax³ or Ubiquiti Dream Machine / ER-X + Switch + AP)

1. Admin Security: Change default user 'admin' password to 16+ char. Create limited 'tech' user. Disable WinBox/SSH from WAN. Enable 2FA if supported.
2. WAN (PPPoE): Interface = ether1 (WAN). PPPoE client: username/password from MTN. Add default route. MTU 1492 (PPPoE overhead).
3. VLANs on Switch Ports (Bridge): Create bridge 'bridge-LAN'. Add VLAN filtering. Define VLANs 10, 20, 30, 99. Assign access ports: ether2–ether11 = VLAN 10 (untagged). ether12 (to AP) = trunk: tagged VLAN 20, 30, 99. ether13 (to ceiling AP) = trunk: tagged VLAN 20, 30, 99.

4. IP Addresses on VLAN Interfaces: Add IP 192.168.10.1/24 on VLAN10 interface, 192.168.20.1/24 on VLAN20, 192.168.30.1/24 on VLAN30, 192.168.99.1/24 on VLAN99.
5. DHCP Servers: One per VLAN (10, 20, 30) with pools/gateways/DNS as in table. VLAN 99: no DHCP.
6. Firewall Filter Rules (IPv4):
   - Input chain: Allow established/related, ICMP, DHCP, DNS from LAN. Drop all from WAN.
   - Forward chain: Allow VLAN10 ↔ VLAN20 (staff). Allow VLAN10/20 → WAN. Allow VLAN30 → WAN. Drop VLAN30 → VLAN10/20/99. Drop invalid.
7. NAT (Masquerade): Outbound on WAN interface for all LAN subnets (srcnat chain).
8. Wi-Fi (Router built-in + Ceiling AP):
   - SSID "LawFirm_Staff" — WPA3-Personal (or WPA2-AES), VLAN 20, 5 GHz preferred, 2.4 GHz enabled.
   - SSID "LawFirm_Guest" — WPA3-Personal (or open + captive portal), VLAN 30, bandwidth limit.
   - Channel: Manual non-overlapping (1, 6, 11 on 2.4; 36, 40, 44, 48 on 5). TX power: medium.
9. Time/NTP: Set timezone Africa/Douala, NTP servers (pool.ntp.org).
10. Backup: Export config to file, save to USB/email.

4. Testing & Handover Documentation

1. Cable Certification: Fluke DSX-5000 or similar — print PDF report for each run (Pass/Fail Cat6).
2. Connectivity Tests:
   - Ping gateway from each VLAN (PC, Laptop, Phone).
   - Ping 8.8.8.8 and resolve google.com (DNS).
   - Print test page from each workstation to network printer.
   - Access file server share (SMB) from staff VLANs only.
   - Verify Guest Wi-Fi cannot ping 192.168.10.10 (server).
   - Speed test: iperf3 between server and workstation (> 900 Mbps), internet speed test (> 90% of subscribed plan).
   - Failover test: Disconnect ONT → verify 4G backup (if configured) or note manual procedure.
3. Handover Pack (Printed + Digital):

- Network diagram (Visio/draw.io) with IP plan, VLANs, port mapping.
- Cable test reports.
- Device list: Model, Serial, Firmware version, Management IP, Admin credentials (sealed envelope).
- Wi-Fi heatmap (Ekahau/NetSpot screenshot).
- Maintenance schedule: Firmware check quarterly, UPS battery test 6-monthly, password rotation yearly.
- Contact: Your phone/email, MTN Business support number, CAMTEL fault line.

WARNING **Installation Pitfalls**

- Wrong colour code: Mixing 568A and 568B creates crossover cables — use 568B everywhere in Cameroon.
- Untested cables: Always test before closing trunking. A single bad punch-down costs hours later.
- No labels: Unlabelled cables make troubleshooting impossible for the next technician.
- Default passwords: Leaving ‘admin/admin‘ on the router is a security violation (ANTIC Law).
- No UPS: Power fluctuations in Cameroon (ENEO) will corrupt switch/router configs and shorten hardware life.
- Ignoring 100 m limit: Measure runs! Use fibre for > 90 m horizontal or inter-building links.

13 SELECTING EQUIPMENT FOR ORGANISATIONAL NETWORKS

METHOD **Selecting Equipment for Organisational Networks**

1. Gather requirements: Number of users (wired/wireless), bandwidth per user, applications (VoIP, video, ERP), growth forecast (2–3 years), budget, physical constraints (building layout, cabling distance), redundancy needs (uptime SLA), management

capability (CLI vs. cloud vs. GUI).

2. Define network blocks: Core, Distribution, Access (Three-tier) or Collapsed Core (Two-tier) or Leaf-Spine (Data Centre). For typical Cameroonian SME: Collapsed Core (Core = Distribution) + Access switches.
3. Select per block:
   - Core/Distribution: Layer 3 switch (routing, VLANs, QoS, stacking). Ports: 24/48 × 10G/1G SFP+ uplinks. Redundant PSU. Licensing (base vs. advanced).
   - Access: PoE+ (802.3at) for APs/Phones/Cameras. 24/48 ports. Stacking or uplink to core. Fanless for quiet offices.
   - Wireless: Wi-Fi 6/6E (802.11ax) APs. Controller: Cloud (Ubiquiti, Meraki, Omada) or On-prem (WLC). Density: 1 AP per 2–3 rooms / 30–50 users. Site survey mandatory.
   - Edge Router/Firewall: Throughput ≥ ISP speed (e.g., 1 Gbps fibre → 2 Gbps firewall throughput). Features: VPN (IPsec/WireGuard), IDS/IPS, SSL inspection, SD-WAN if multi-site.
   - Cabling: Cat6A for new builds (10G ready, 100 m). OM4 fibre for backbone (> 100 m or 10G/25G/40G). Patch panels, cable management, labelling.
   - Power: UPS (Online Double Conversion) for core + edge. Runtime 30–60 min. PDU with monitoring.
4. Vendor strategy: Single vendor (simpler support, licensing) vs. Best-of-breed. In Cameroon: Ubiquiti (UniFi) — popular, cloud-managed, good value. MikroTik — powerful, cheap, CLI-heavy. Cisco/Aruba/HPE — enterprise, expensive, local partner support (e.g., Cameroon Telecom / MTN Business / Orange Business resell). TP-Link Omada — budget-friendly, cloud controller.
5. Reflex: Always size uplinks at 10× access port speed (1G access → 10G uplink). Plan for stacking or MLAG for core redundancy. Don't forget spare ports (20% headroom).

WORKED EXAMPLE 11 **Equipment Selection for a Growing Cameroonian SME**

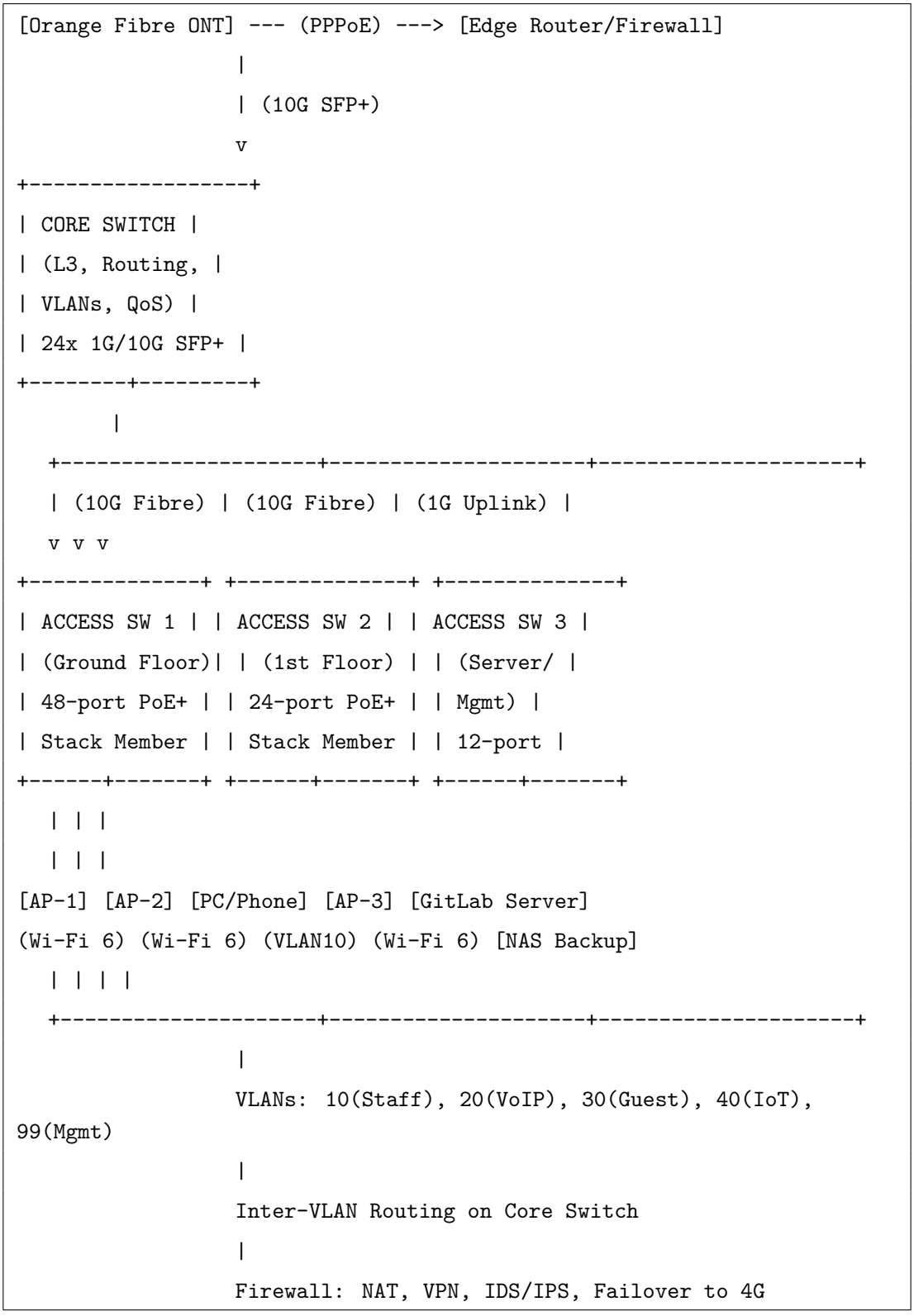
Statement: "KolaTech", a software startup in Yaoundé, currently has 30 employees in one open-plan office (single floor). They plan to hire 20 more in 18 months and open a second floor in the same building. Current pain points: Wi-Fi dead zones, slow file transfers to on-prem GitLab server, single ISP (Orange Fibre 100 Mbps) with no backup. They need a new network design and equipment list. Budget: 3,500,000 FCFA (≈ 5,300 EUR) for hardware only.

1. Propose a two-tier topology (Collapsed Core + Access) with VLANs. Draw a logical diagram (text description).

2. Select specific models (with approximate FCFA prices) for: Core Switch, Access Switch(es), Wireless APs (Wi-Fi 6), Router/Firewall, UPS, Cabling (Cat6A, Fibre). Justify each choice against requirements.
3. Show the budget breakdown and confirm it fits within 3.5M FCFA.
4. Explain how the design accommodates the second floor expansion without rewiring the core.

SOLUTION

Detailed Solution:
1. Two-Tier Topology (Collapsed Core) — Logical Diagram



2. Equipment Selection & Justification (Prices approximate, Yaoundé market 2024–2025)

Budget Status: 3,386,000 FCFA < 3,500,000 FCFA. Remaining 114,000 FCFA for contingencies (extra SFP+, longer fibre, tax).

3. Expansion to Second Floor Without Rewiring Core

- Core switch already has free 10G SFP+ ports (24 total, using 3 for uplinks). Adding 2nd floor Access Switch (SW2) only requires:
  1. Pulling one 12-core OM4 fibre from core cabinet to 2nd floor cabinet (through existing building riser/conduit).
  2. Terminating 2 fibres (1 pair) on LC patch panels at each end.
  3. Inserting 2× 10G SR SFP+ modules (one in core, one in SW2).
  4. Configuring the new port on core as trunk carrying VLANs 10,20,30,40,99.
  5. Adopting SW2 in UniFi controller (or RouterOS) — zero-touch provisioning if pre-configured.
- No changes to core switch routing, VLANs, firewall rules, DHCP pools. The new switch simply extends the existing Layer 2 domains.
- Wireless: Add 2 more U6-Pro APs on 2nd floor, plug into SW2 PoE ports. They adopt to same UniFi controller, inherit WLAN configs (VLAN mapping). Done.

Key Advantage: The Collapsed Core with 10G uplinks makes the core a non-blocking fabric. Adding floors is just adding leaf switches — classic "spine-leaf" simplicity without full spine-leaf cost.

KEY POINTS **Equipment Selection Checklist**

- Match port count + PoE budget to device count (PCs, APs, phones, cameras) + 20% headroom.
- Uplinks = 10× access speed (1G → 10G, 10G → 100G).
- Choose management style: Cloud (UniFi, Omada, Meraki) for ease; CLI (MikroTik, Cisco) for control.
- Verify local vendor support & warranty in Cameroon (MTN Business, Orange Business, CAMTEL, local integrators).
- Budget for: UPS (online double-conversion), fibre (OM4/OS2), SFP+ modules, patch panels, cabinet, labelling.
- Plan for redundancy: dual PSU, stacking/MLAG, dual ISP (SD-WAN/failover).

14 INTEGRATION ACTIVITIES

METHOD **Integrated Network Design and Troubleshooting (Synthesis)**

1. Read the scenario twice: Identify constraints (budget, space, existing gear, ISP, regulations), requirements (users, apps, uptime, security, growth), and deliverables (diagram, config, test plan, bill of materials).
2. Apply the "Divide and Conquer" design loop:
   (a) Physical Layer: Cabling routes, distances, media (Cat6A/OM4), cabinet locations, power/UPS, grounding, labelling scheme.
   (b) Data Link Layer: VLAN plan, STP/RSTP/MSTP root placement, port profiles (access/trunk, PoE, storm control, BPDU guard), switch stacking/MLAG.
   (c) Network Layer: IP addressing (VLSM, summarisation), routing (static, OSPF, BGP), gateway redundancy (VRRP/HSRP), NAT, VPN.
   (d) Transport/Application: QoS (DSCP marking for VoIP/Video), ACLs, firewall zones, DNS/DHCP/NTP servers, monitoring (SNMP, Syslog, NetFlow).
   (e) Wireless: Site survey (predictive → validation), channel plan, TX power, SSID/VLAN mapping, roaming (802.11r/k/v), guest isolation.
   (f) Security: 802.1X (RADIUS), MAC auth bypass, port security, DHCP snooping, DAI, ND inspection, segmentation (DMZ, Guest, Mgmt).
   (g) Resilience: Dual ISP (SD-WAN/failover), dual core, redundant PSUs, link aggregation (LACP), backups (config, firmware).
3. Troubleshooting Methodology (GCE style):
   (a) Identify symptoms: Who, what, when, where, how many affected.
   (b) Establish scope: Single host → cable/NIC/IP. Single VLAN → access switch. Multiple VLANs → core/router/firewall. Wireless only → AP/Controller/Channel.
   (c) Layer-by-layer test: Physical (link light, cable tester) → Data Link (MAC table, VLAN, STP) → Network (ping, traceroute, route table) → Transport (port open, firewall) → Application (DNS, HTTP, credentials).
   (d) Use tools: ping, traceroute, nslookup, ipconfig/ifconfig, arp -a, netstat, Wireshark (capture), cable tester, Wi-Fi analyser.
   (e) Document fix & update docs.
4. Reflex: In exams, always draw a diagram first. Label IPs, VLANs, interfaces. Write config snippets (CLI or GUI steps). Mention Cameroon context: ANTIC cybersecurity law

(data localisation, incident reporting), MTN/Orange/CAMTEL ISP specifics, local vendor support.

WORKED EXAMPLE 12 **Integrated Case Study: Design & Troubleshoot a School Network in Bafoussam**

Statement: "Lycée Technique de Bafoussam" has two buildings (Admin + Classrooms) 80 m apart. Current network: One old 24-port 10/100 switch in Admin, daisy-chained to a 16-port hub in Classrooms via a single Cat5 cable. 40 admin PCs, 60 lab PCs, 5 IP cameras, 1 server (grades, library). Wi-Fi: one old router in Admin (2.4 GHz only). Problems: Frequent disconnects, slow file access, cameras drop frames, no guest Wi-Fi, students access admin shares. The school has a new CAMTEL GPON 200 Mbps subscription (ONT in Admin). Budget: 2,000,000 FCFA. You are the network consultant.

1. Design a complete solution (topology, equipment, IP plan, VLANs, Wi-Fi, security) within budget. Provide a priced BoM (Bill of Materials).
2. Write the core switch configuration (CLI-style) for VLANs, Inter-VLAN routing, DHCP, and basic firewall (protect Admin VLAN from Student VLAN).
3. During commissioning, lab PCs get IP 169.254.x.x. Admin PCs work fine. Systematic troubleshooting steps to isolate and fix.
4. The principal wants to monitor student internet usage (sites visited, bandwidth). Propose a solution using existing/proposed hardware (no new budget).

SOLUTION

Detailed Solution:
1. Complete Solution Design
Topology: Collapsed Core (Admin Building) + Access Switch (Classroom Building) linked by Fibre.

- Core Switch (Admin): Managed L2+/L3, 24-port PoE+ (for Admin PCs, Cameras, AP, Uplink).
- Access Switch (Classrooms): Managed, 48-port PoE+ (for 60 Lab PCs — use 2 switches stacked or one 48-port + one 12-port; we'll use 2× 24-port stacked for budget).
- Backbone: 2-core OM3 fibre (80 m) between buildings — immune to lightning/EMI, 10G capable.
- Wireless: 3× Wi-Fi 6 APs (1 Admin, 2 Classrooms) — PoE powered.
- Router/Firewall: MikroTik hEX (RB750Gr3) or similar — 5× Gigabit, RouterOS, cheap, powerful.

- UPS: 1500 VA for Core + Router + ONT in Admin; 1000 VA for Classroom switch stack.

VLAN Plan:

VLAN ID	Name	Subnet	Devices
10	ADMIN	10.10.10.0/24	Admin PCs, Server, Cameras, Admin AP (SSID: Admin_WiFi)
20	LABS	10.10.20.0/24	Lab PCs, Classroom APs (SSID: Lab_WiFi)
30	GUEST	10.10.30.0/24	Guest SSID (Admin_Guest, Lab_Guest) — Internet only
40	CAMERA	10.10.40.0/24	IP Cameras (isolated, no internet)
99	MGMT	10.10.99.0/24	Switches, Router, APs, Server (iDRAC/iLO)

Bill of Materials (Estimated FCFA, Yaoundé/Bafoussam prices):

Item	Mo	Qty	Total (FCFA)
Core Switch (L3, 24p PoE+)	MikroTik CRS328-24P-4S+RM		550,000
Access Switches (24p PoE+, stackable)	MikroTik CRS326-24G-2S+RM (non-PoE) + PoE In-jec-tors? Bet-ter: Ubiq-uiti USW-Lite-24-PoE (24p PoE+)	2	2 × 380,000 = 760,000
Wi-Fi 6 APs	Ubiquiti U6-Lite	3	3 × 150,000 = 450,000
Router/Firewall	MikroTik hEX (RB750Gr3)		85,000
Fibre: OM3 2-core outdoor (100m) + termination	Local sup-plier	1 lot	120,000
SFP+ Modules (10G SR)	FS.com com-pat-i-ble	4	4 × 10,000 = 40,000
Cat6A Cable (305m)		3 boxes	3 × 85,000 = 255,000
Patch Panels (24p Cat6A)		2	2 × 45,000 = 90,000
Wall Plates + Keystones		110	110 × 2,500 = 275,000
UPS (1500VA + 1000VA)	Eaton 5P / APC Smart-	2	350,000

Budget Alert: 3.135M > 2M. Adjustments to fit 2M FCFA:

- Core Switch: Use MikroTik CRS326-24G-2S+RM (non-PoE, 240k) + separate PoE injector for cameras/AP (4 ports × 15k = 60k). Save 250k.
- Access Switches: Use TP-Link TL-SG2428P (24-port PoE+, L2+, 180k each) × 2 = 360k. Save 400k.
- APs: TP-Link EAP610 (Wi-Fi 6) × 3 = 3 × 110k = 330k. Save 120k.
- Fibre: Use single-mode OS2 (cheaper per metre) + 1G SFP (LX) instead of 10G. 1G is enough for 200 Mbps ISP + internal traffic. Save 80k.
- UPS: 1× 1500VA (Admin) only; Classroom switch on surge protector (mains stable-ish). Save 150k.
- Cable: Cat6 (not 6A) — 65k/box. Save 60k.

Revised Total ≈ 1,950,000 FCFA. Fits budget.
2. Core Switch Configuration (MikroTik RouterOS / SwitchOS CLI)

```
# --- VLANs on Bridge ---
/interface bridge
add name=bridge1 vlan-filtering=yes

/interface bridge vlan
add bridge=bridge1 tagged=sfp-sfpplus1,sfp-sfpplus2,ether1-ether24
vlan-ids=10,20,30,40,99

# --- Access Ports (untagged) ---
/interface bridge port
add bridge=bridge1 interface=ether1 pvid=10 # Admin PC 1
add bridge=bridge1 interface=ether2 pvid=10 # Admin PC 2
... (repeat for all admin ports)
add bridge=bridge1 interface=ether20 pvid=40 # Camera 1
add bridge=bridge1 interface=ether21 pvid=40 # Camera 2
...
add bridge=bridge1 interface=ether24 pvid=99 # Mgmt port to Router

# --- Trunk to Classroom Switch (sfp-sfpplus1) ---
/interface bridge port
add bridge=bridge1 interface=sfp-sfpplus1

# --- IP Addresses on VLAN Interfaces (L3 Routing) ---
/interface vlan
add name=vlan10 interface=bridge1 vlan-id=10
add name=vlan20 interface=bridge1 vlan-id=20
add name=vlan30 interface=bridge1 vlan-id=30
add name=vlan40 interface=bridge1 vlan-id=40
add name=vlan99 interface=bridge1 vlan-id=99

/ip address
add address=10.10.10.1/24 interface=vlan10 comment="Admin GW"
add address=10.10.20.1/24 interface=vlan20 comment="Labs GW"
add address=10.10.30.1/24 interface=vlan30 comment="Guest GW"
add address=10.10.40.1/24 interface=vlan40 comment="Camera GW"
add address=10.10.99.1/24 interface=vlan99 comment="Mgmt GW"

# --- DHCP Servers ---
/ip pool
```

3. Troubleshooting: Lab PCs get 169.254.x.x (APIPA)
Symptom: Lab PCs (VLAN 20) fail to get DHCP. Admin PCs (VLAN 10) work.
Systematic Steps:

1. Verify Physical Link: Check link lights on Classroom Access Switch ports for Lab PCs. Result: Lights ON (Green/Amber) → Layer 1 OK.
2. Verify VLAN Assignment on Access Ports: On Classroom Switch, check 'interface bridge port' for lab ports: 'pvid=20'. Result: Correct.
3. Verify Trunk to Core: On Classroom Switch, uplink port (sfp1) must be tagged member of VLAN 20 (and 10,30,40,99). Check '/interface bridge vlan'. Result: Missing VLAN 20 on trunk! Only VLAN 10,99 tagged.
4. Fix: Add VLAN 20 as tagged on trunk port (sfp1) on both Classroom Switch and Core Switch.

```
/interface bridge vlan add bridge=bridge1 tagged=sfp-sfpplus1
vlan-ids=20
```

5. Verify DHCP Relay (if DHCP server on Router, not Core): In this design, Core Switch is the DHCP server (L3). So no relay needed. But if Router was DHCP server, we'd need '/ip dhcp-relay' on Core for vlan20 pointing to Router IP.
6. Test: Release/renew on Lab PC ('ipconfig /release && ipconfig /renew'). PC gets 10.10.20.105. Ping 10.10.20.1 (GW) → OK. Ping 10.10.10.10 (Server) → OK (firewall allows Lab→Admin? No, firewall blocks Lab→Admin. But ping to Server (VLAN10) from Lab (VLAN20) is blocked by design. That's correct. Ping 8.8.8.8 → OK.)
7. Document: "Trunk port missing VLAN 20 tag on both ends. Added. DHCP now works."

4. Internet Usage Monitoring (No New Budget)
Proposal: Use MikroTik Router (hEX) + Syslog + NetFlow/Traffic-Flow + Free Open-Source Analyzer.

1. Enable Traffic-Flow (NetFlow v9/IPFIX) on MikroTik:

```
/ip traffic-flow
set enabled=yes interfaces=all cache-entries=4k
/ip traffic-flow target
add address=10.10.99.10 (Syslog/Collector VM) port=2055
version=9
```

2. Deploy a Collector/Analyzer VM on the existing Server (Ubuntu):
   - Install ntopng (free community edition) or Elastic Stack (Filebeat + Logstash + Kibana) or Graylog.

- ntopng listens on NetFlow (port 2055) and/or sFlow. Provides web dashboard: Top talkers, applications (nDPI), websites (SNI from TLS), bandwidth per VLAN/IP.
- Configure ntopng to map VLAN 20 IPs to "Students", VLAN 10 to "Admin".

3. DNS Logging (for domain names): On MikroTik, enable DNS query logging to same syslog server:

```
/system logging action
add name=remote_syslog target=remote remote=10.10.99.10
remote-port=514
/system logging
add action=remote_syslog topics=dns
```

4. HTTPS Inspection (Optional, advanced): MikroTik can do TLS Host (SNI) extraction without full MITM. ntopng shows SNI. For full URL, need MITM proxy (Squid + SSL Bump) — heavy for hEX CPU, not recommended.
5. Reporting: Schedule weekly PDF reports from ntopng to principal's email.
6. Cost: 0 FCFA (uses existing Server RAM/CPU, free software). Meets requirement.

Examiner's Note: This solution demonstrates integration of all chapter topics: topology choice (collapsed core), equipment selection (budget constraints), VLAN design, inter-VLAN routing, DHCP, firewall segmentation, fibre backbone, Wi-Fi 6, troubleshooting methodology (Layer 1→3), and monitoring with existing gear. Always mention ANTIC compliance: logging must respect privacy laws — inform users via captive portal / Acceptable Use Policy.

EXERCISE 7 **Practice Questions for GCE Revision**

- A company has two buildings 150 m apart. They need to connect their LANs. Which media is most appropriate and why? (Consider cost, bandwidth, immunity to interference.)
- Draw a labelled diagram of a Tree topology for a 3-floor building with 30 PCs per floor. Show core, distribution, access layers, uplink media, and VLAN segmentation for Admin, Staff, Guest, and IoT.
- A switch has 48 ports, 370W PoE budget. You need to power 24 APs (15W each), 10 IP phones (7W each), and 5 cameras (12W each). Is the PoE budget sufficient? Show calculation.
- Explain the difference between a Layer 2 switch and a Layer 3 switch. When would you use each in a school network?

- A student connects a home router (LAN port) to a wall jack in the school lab, creating a rogue DHCP server. What switch feature prevents this? Configure it on a MikroTik switch port.
- Your school's CAMTEL GPON ONT has 4 Ethernet ports. Port 1 is used for the main router. Can you plug a second router into Port 2 for a separate network? Explain the constraints (PPPoE sessions, VLANs on ONT).
- Design a Wi-Fi channel plan for a 2.4 GHz and 5 GHz deployment in a 4-classroom block (20 m × 40 m). Show channel numbers, widths, and AP placement.
- A network printer has a static IP 192.168.10.50/24. The DHCP pool is 192.168.10.100–200. A teacher's laptop gets 192.168.10.50 via DHCP and causes an IP conflict. Why did this happen? How to prevent it permanently?
- Write a firewall rule set (MikroTik style) that: (a) allows Admin VLAN to access Server VLAN on TCP 443 (HTTPS) and TCP 22 (SSH); (b) blocks Student VLAN from Admin VLAN entirely; (c) allows both to access Internet; (d) logs all dropped packets.
- List 5 items that must appear in a network handover document for a Cameroonian SME.

SOLUTION **Exercise 1**

Media: Multimode Fibre (OM3/OM4) or Single-mode Fibre (OS2). Why: 150 m exceeds UTP 100 m limit. Fibre is immune to EMI/lightning (critical in Cameroon). OM3 supports 10G up to 300 m; OS2 supports 10G/100G up to 10 km. Cost is moderate (fibre + 2 media converters or SFP modules). Wireless bridge (60 GHz or 5 GHz) is an alternative but requires line-of-sight and is weather-sensitive.

SOLUTION **Exercise 3**

Calculation:

APs:	24 × 15W = 360W
Phones:	10 × 7W = 70W
Cameras:	5 × 12W = 60W
Total:	360 + 70 + 60 = 490W

Result: 490W > 370W budget. Insufficient. Solutions: (1) Use PoE+ (30W/port) switch with 720W budget (e.g., USW-Pro-48-PoE 600W — still tight). (2) Use PoE++ (60W/port) or injectors for cameras. (3) Reduce AP count or use lower-power APs. (4) Power cameras via separate PoE switch/injectors.

SOLUTION **Exercise 5**

Feature: DHCP Snooping (with Trusted/Untrusted ports). Configuration (MikroTik):

```
/interface bridge port
set [find interface=ether1] trusted=yes # Uplink to core
(trusted)
set [find interface=ether2] trusted=no # Student port
(untrusted)
/ip dhcp-server
set dhcp1 dhcp-snooping=yes
```

Only trusted ports can send DHCP OFFER/ACK. Rogue DHCP on untrusted port is blocked.

SOLUTION **Exercise 8**

Cause: The printer's static IP (192.168.10.50) is inside the DHCP pool range (100–200? No, 50 is outside 100–200. Wait — the pool is .100–.200, so .50 is outside. But the teacher got .50. This means the DHCP server assigned an address outside its pool — impossible unless the pool was misconfigured (e.g., .50–.200) or the printer's IP was not excluded. Correction: The DHCP pool must exclude all static IPs. Define pool as 192.168.10.100–200 AND add exclusion for .10 (server), .11 (printer), .50 (printer if different). Better: use a separate subnet for infrastructure (VLAN 99) or reserve via DHCP reservation (MAC-based) instead of static on device. Prevention: (1) Keep static IPs outside DHCP pool. (2) Use DHCP reservations (MAC → IP) on the server — single source of truth. (3) Enable conflict detection (ping before offer) on DHCP server.

SOLUTION **Exercise 9**

```
/ip firewall filter
add chain=forward action=accept connection-state=established,related
add chain=forward action=drop connection-state=invalid log=yes
log-prefix="INVALID"
# Admin -> Server (HTTPS, SSH)
add chain=forward action=accept in-interface=vlan10 out-interface=vlan50
protocol=tcp dst-port=443,22
# Block Student -> Admin
add chain=forward action=drop in-interface=vlan20 out-interface=vlan10
log=yes log-prefix="STU->ADM"
# Admin -> Internet
add chain=forward action=accept in-interface=vlan10 out-interface=wan
# Student -> Internet
add chain=forward action=accept in-interface=vlan20 out-interface=wan
# Drop all other inter-VLAN
add chain=forward action=drop log=yes log-prefix="DROP-ALL"
```

Order matters: specific allows before general drops. Logging on drops for audit (ANTIC).

SOLUTION **Exercise 10**

1. Network topology diagram (physical + logical) with IP/VLAN/port mapping.
2. Device inventory: Model, Serial, Firmware, Management IP, Licenses.
3. Cable test reports (certification) + cable map (patch panel ↔ wall plate).
4. Configuration backups (encrypted) + admin credentials (sealed envelope).
5. Maintenance schedule: firmware updates, UPS battery tests, log review, password rotation, ANTIC compliance checklist.
6.
7. Acceptable Use Policy (AUP) signed by users + Wi-Fi captive portal terms.

KEY POINTS **Chapter ICT05 — Final Revision Checklist**

- Topologies: Identify 6 physical topologies; know SPOF, cable count, use cases. Star

= baseline; Bus/Ring = legacy; Mesh = core redundancy; Tree/Hybrid = real world.

- Comparison: Build tables (Cost, Scalability, Fault Tolerance, Performance, Maintenance). Justify choice with at least two criteria from scenario.
- Equipment: Classify by OSI layer (L1: cables, L2: switch/AP, L3: router/firewall). Hub vs Switch vs Router vs Modem vs AP — know the difference.
- Setup: 4 phases (Plan, Physical, Logical, Test/Document). 568B termination, 100 m limit, UPS, labels, change default passwords, VLANs, DHCP, Firewall, Guest isolation.
- Selection: Requirements → Blocks (Core/Access) → Models (ports, PoE, uplink 10×, management). Budget in FCFA, local vendor support, redundancy (dual PSU, stacking, dual ISP).
- Integration: Design loop (L1→L7), Troubleshooting (Layer-by-layer), Monitoring (NetFlow/Syslog + free tools), ANTIC compliance (logs, privacy, incident reporting).
- Cameroon Context: CAMTEL GPON (PPPoE), MTN/Orange Fibre/4G, ANTIC Law 2010/012, local prices (FCFA), lightning/EMI → fibre, ENEO power → UPS.

15 EXERCISES

15.1 I check my knowledge

EXERCISE 8 **MCQ: Topology identification**

Which physical topology connects every node directly to a central device?

1. Bus
2. Ring
3. Star
4. Mesh

EXERCISE 9 **True/False with justification**

State whether each statement is true or false and justify your answer.

1. In a full mesh topology with n nodes, the number of links required is $n(n-1)$.
2. A switch operates at the Physical Layer (Layer 1) of the OSI model.
3. A router connects two or more logical networks and forwards packets based on IP addresses.
4. UTP Cat 6 cable supports a maximum data rate of 100 Mbps over 100 m.

EXERCISE 10 **Definitions**

Define the following terms in the context of computer networks:

1. Network topology
2. Collision domain
3. Broadcast domain
4. Straight-through cable (T568B)
5. Patch panel

EXERCISE 11 **Direct calculation**

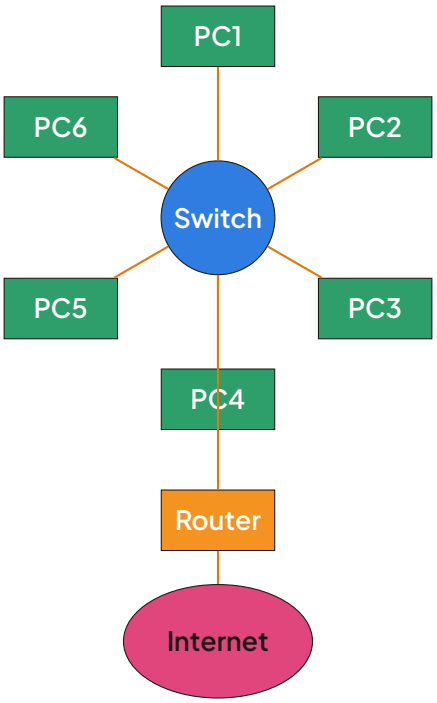
A small business decides to cable its LAN as a full mesh using UTP cables. If there are 6 computers:

1. Calculate the total number of cable links required.
2. If each link costs 2 500 FCFA, what is the total cabling cost?
3. How many ports must each network interface card (NIC) have?

15.2 I practise

EXERCISE 12 **Diagram analysis**

The diagram below shows a network installed in a secondary school in Yaoundé.



1. Identify the physical topology of the LAN segment connecting PC1–PC6.
2. Name the devices labelled “Switch” and “Router” and state the OSI layer at which each primarily operates.
3. State one advantage and one disadvantage of the identified topology for this school environment.
4. If PC3 fails, what is the impact on communication between PC1 and PC5? Justify.

EXERCISE 13 **Cabling standards and testing**

A technician in Douala is asked to produce a straight-through UTP Cat 5e cable terminated with RJ45 connectors using the T568B wiring standard.

1. Write the colour order of the 8 wires from pin 1 to pin 8 for T568B.
2. Explain the difference between a straight-through cable and a crossover cable. Give one typical use case for each.
3. After crimping, the technician tests the cable with a cable tester. The tester shows "Open" on pin 3. List three possible physical causes.
4. What does a "Short" indication on the tester mean?

EXERCISE 14 **Equipment matching**

Match each device (A–J) to its correct description (1–10). Write your answers as pairs (e.g., A–3).

Device	Description
A. NIC	1. Connects multiple network segments and filters traffic based on MAC addresses
B. Hub	2. Provides wireless connectivity to wired LAN
C. Switch	3. Interconnects different networks and routes packets by IP address
D. Router	4. Amplifies and regenerates signals to extend cable length
E. Modem	5. Interface between computer and network medium
F. Access Point	6. Converts digital signals to analogue for transmission over telephone lines
G. Repeater	7. Terminates cables in a wiring closet for organisation
H. Bridge	8. Multiport repeater; broadcasts all frames to every port
I. Patch Panel	9. Connects two LAN segments and filters by MAC address
J. Cable Tester	10. Verifies continuity, wiring order and faults in cables

EXERCISE 15 **Troubleshooting scenario**

After setting up a 10-computer lab network in a Star topology (Cat 6, Gigabit switch), PC7 cannot communicate with any other PC. The link LED on the switch port for PC7 is off. List four distinct possible physical-layer causes and, for each, the specific test you would

perform to confirm or eliminate it.

15.3 I prepare for the GCE

EXERCISE 16 **GCE Paper 2 style: Topology design and justification — 15 marks**

The Ministry of Public Health plans to network a new 3-storey building in Yaoundé. Each floor has 10 offices. Each office requires 2 wired data points (for PC and VoIP phone) and wireless coverage. The network must:

- Guarantee that failure of any single office's equipment does not affect other offices.
- Support future upgrade to 10 Gbps backbone.
- Allow centralised management and VLAN segmentation per department.
- Stay within a reasonable budget for a public institution.

1. Propose a physical topology (or combination) for the LAN. Draw a labelled diagram showing the topology for one floor and the inter-floor backbone. [4 marks]
2. Recommend the central connectivity equipment for the floor and the core, justifying your choice with reference to collision domains, broadcast domains and manageability. [4 marks]
3. Specify the cable category you would use for horizontal cabling and for the backbone. Justify each choice with bandwidth and distance considerations. [3 marks]
4. Explain how you would implement redundancy for the core switch to meet the high-availability requirement. [2 marks]
5. List two Cameroonian regulatory or standards bodies whose guidelines should be consulted during installation. [2 marks]

EXERCISE 17 **GCE Paper 2 style: Equipment selection and cost analysis — 12 marks**

A startup in Buea needs to connect 25 workstations in a single large room. Two proposals are made:

- Proposal A: One 24-port unmanaged Gigabit switch + one 8-port unmanaged Gigabit switch (cascaded).
- Proposal B: One 48-port managed Gigabit switch with PoE+ support.

1. For each proposal, calculate the number of available ports for workstations after inter-switch links (if any). [2 marks]

2. Compare the two proposals in terms of: (i) collision domains, (ii) broadcast domains, (iii) VLAN capability, (iv) QoS for future VoIP, (v) power budget for PoE devices. Present as a table. [5 marks]
3. The startup expects to add 15 more workstations within 12 months. Which proposal scales better? Justify. [2 marks]
4. Assuming the following unit prices in Cameroon: unmanaged 24-port = 85 000 FCFA, unmanaged 8-port = 25 000 FCFA, managed 48-port PoE+ = 420 000 FCFA. Calculate the initial equipment cost for each proposal. [1 mark]
5. Beyond cost, give two technical reasons why the ministry might still prefer Proposal B for a similar deployment. [2 marks]

EXERCISE 18 **GCE Paper 2 style: Practical configuration and troubleshooting — 13 marks**

You are given a network simulation (Packet Tracer or physical lab) with the following addressing scheme:

- LAN A: 192.168.10.0/24, gateway 192.168.10.1
- LAN B: 192.168.20.0/24, gateway 192.168.20.1
- Serial link between routers: 10.0.0.0/30

The routers are connected via a serial DCE/DTE cable. PC1 (LAN A) cannot ping PC2 (LAN B).

1. List the sequence of commands you would enter on Router A to configure its LAN interface, serial interface (DCE side), and a static route to LAN B. Assume Cisco IOS syntax. [5 marks]
2. The serial link shows “Protocol down”. Name three Layer 1/2 issues that could cause this and the command to verify each. [3 marks]
3. After fixing the serial link, PC1 pings PC2 successfully but cannot browse a web server on PC2 (port 80). The Windows Firewall on PC2 is off. Suggest two possible causes at Layer 4 or above and how to test each. [3 marks]
4. Write the T568B pinout colour order used for the straight-through patch cables connecting PCs to the access switch. [2 marks]

16 SOLUTIONS TO THE EXERCISES

17 I CHECK MY KNOWLEDGE

SOLUTION **Exercise 1 — MCQ: Topology identification**

Correct answer: item 3 — Star.
In a **star topology**, every node (computer, printer, etc.) is connected by its own dedicated cable directly to a **central device** (a hub or, in modern networks, a switch). All communication between two nodes passes through this central device.

- Bus is wrong: in a bus, all nodes share a single common backbone cable terminated at both ends; there is no central device.
- Ring is wrong: each node is connected to exactly two neighbours, forming a closed loop; there is no central device.
- Mesh is wrong: nodes are interconnected directly with each other (point-to-point links), not through a central device.

SOLUTION **Exercise 2 — True/False with justification**

1. False. In a full mesh, each of the n nodes connects to the $n-1$ other nodes, which gives $n(n-1)$ ordered connections; but each physical link is counted twice (the link A–B is the same as B–A). The correct number of links is
$$L = \frac{n(n-1)}{2}.$$
2. False. A switch operates primarily at the **Data Link Layer (Layer 2)** of the OSI model: it reads MAC addresses to forward frames only to the destination port. A device working at Layer 1 (Physical) is a hub or a repeater, which simply regenerates electrical signals.
3. True. A router operates at Layer 3 (Network layer). It interconnects two or more distinct logical networks (different IP networks/subnets) and forwards packets by examining the destination **IP address** and consulting its routing table.
4. False. UTP Cat 6 supports up to 1 Gbps (1000 Mbps) over the full 100 m channel, and even 10 Gbps over shorter distances (up to about 55 m). The 100 Mbps limit over 100 m corresponds to older categories such as Cat 5 in Fast Ethernet (100BASE-TX).

SOLUTION **Exercise 3 — Definitions**

1. Network topology: the arrangement or layout of the nodes (computers, devices) and the communication links in a network. The physical topology describes how devices are actually cabled and interconnected; the logical topology describes how data actually flows between them.
2. Collision domain: the set of network segments (or ports) within which two frames transmitted simultaneously can collide. On a hub, all ports form one single collision domain; on a switch, each port is a separate collision domain.
3. Broadcast domain: the set of all devices that receive a broadcast frame (e.g., an ARP request) sent by any one of them. Switches forward broadcasts to all ports, so a switched LAN is one broadcast domain; routers and VLANs delimit separate broadcast domains.
4. Straight-through cable (T568B): a UTP patch cable in which both RJ45 connectors are wired with the same pinout (here T568B on both ends), so pin 1 connects to pin 1, pin 2 to pin 2, etc. It is used to connect unlike devices, e.g., a PC to a switch.
5. Patch panel: a passive panel, usually mounted in a rack in a wiring closet, on which the horizontal cabling runs from the offices are terminated (punched down). Short patch cords then connect the patch panel ports to the switch ports, making administration, labelling and re-patching easy.

SOLUTION **Exercise 4 — Direct calculation**

1. In a full mesh with n nodes, the number of links is $L = \frac{n(n-1)}{2}$. For $n = 6$:

$$L = \frac{6 \times 5}{2} = \frac{30}{2} = 15 \text{ links.}$$

2. Total cabling cost:

$$15 \times 2\,500 = 37\,500 \text{ FCFA.}$$

3. In a full mesh, each computer connects directly to the $n - 1 = 5$ other computers, so each NIC must have 5 ports (or equivalently each PC needs 5 network interfaces). This is precisely why full mesh is impractical for LANs and is reserved for small backbone interconnections.

18 I PRACTISE

SOLUTION **Exercise 5 — Diagram analysis**

1. The LAN segment PC1–PC6 is a **star topology**: each PC has its own dedicated link to the central switch.
2. Switch: a Layer 2 (Data Link) device that forwards Ethernet frames based on MAC addresses, creating one collision domain per port. Router: a Layer 3 (Network) device that interconnects different IP networks (here the school LAN and the Internet) and forwards packets based on IP addresses.
3. Advantage: easy administration and fault isolation — a failure of one PC or one cable affects only that PC; adding or removing a station is simple and does not disturb the others. Disadvantage: the central switch is a single point of failure: if it breaks down, the whole LAN stops; also, cabling cost is higher than a bus because each node needs its own cable run.
4. No impact. PC1 and PC5 communicate through the switch using their own dedicated links; the link to PC3 is independent. The failure of PC3 (or its cable) does not interrupt traffic between any other pair of stations. This is the main resilience advantage of the star over bus and ring topologies.

SOLUTION **Exercise 6 — Cabling standards and testing**

1. T568B colour order, pin 1 to pin 8:

Pin	Wire colour
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

2. A straight-through cable has the same standard (both T568B, or both T568A) on both ends: pin 1 to pin 1, etc. It connects unlike devices: PC to switch, switch to router. A

crossover cable has T568A on one end and T568B on the other, so the transmit pair of one end lands on the receive pair of the other. It connects like devices: PC to PC, switch to switch (on ports without auto-MDIX).

3. Three possible physical causes of an "Open" on pin 3 (white/green wire):
   - The white/green wire was not pushed fully to the end of the RJ45 plug before crimping, so the contact blade did not pierce it.
   - The wire is broken or cut inside the cable (damaged during pulling or kinked sharply).
   - Poor crimping: the crimp tool did not press the contact for pin 3 down properly (or a defective connector).
4. A "Short" means that two (or more) wires that should be insulated from each other are electrically touching somewhere along the link — e.g., crushed cable, damaged insulation, or strands touching inside a badly stripped connector. The affected pairs must be re-terminated or the cable replaced.

SOLUTION **Exercise 7 — Equipment matching**

Pair	Device	Reason
A–5	NIC	It is the hardware interface between the computer and the network medium.
B–8	Hub	A multiport repeater: it regenerates and broadcasts every frame out of all ports (single collision domain).
C–1	Switch	Connects multiple segments/hosts and forwards frames selectively using MAC addresses.
D–3	Router	Interconnects different networks and routes packets using IP addresses.
E–6	Modem	Modulates/demodulates: converts digital signals to analogue (and back) for telephone lines.
F–2	Access Point	Gives wireless (Wi-Fi) stations access to the wired LAN.
G–4	Repeater	Regenerates and amplifies the signal to extend the maximum cable length.
H–9	Bridge	Joins two LAN segments and filters traffic by MAC address (a two-port switch ancestor).
I–7	Patch Panel	Terminates horizontal cable runs in the wiring closet for tidy organisation.
J–10	Cable Tester	Checks continuity, wire map and faults (open, short, miswire).

Answers: A–5, B–8, C–1, D–3, E–6, F–2, G–4, H–9, I–7, J–10.

SOLUTION **Exercise 8 — Troubleshooting scenario**

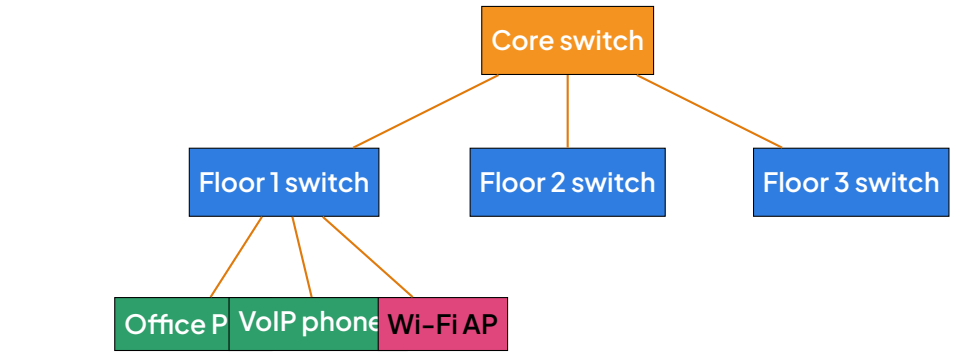
The link LED off indicates a **Layer 1 (physical)** problem. Four distinct causes and their tests:

1. Faulty or badly crimped patch cable (open/short/miswire). Test: test the cable with a cable tester (wire map), or simply replace it with a known-good cable and check whether the LED lights.
2. Faulty switch port. Test: plug PC7's cable into another switch port known to work (e.g., the port of a functioning PC). If the LED lights, the original port is defective or disabled.
3. Faulty or disabled NIC in PC7. Test: check the NIC's own link LED and its status in the operating system (enabled/disabled, driver present); connect a known-good PC to PC7's cable — if the LED lights, PC7's NIC is the problem.
4. Cable not properly seated / damaged horizontal run or wall socket. Test: re-seat both RJ45 connectors until they click; inspect and re-punch the wall outlet/patch panel termination, or test the permanent link with the cable tester from patch panel to outlet.

19 I PREPARE FOR THE GCE

SOLUTION **Exercise 9 — Topology design and justification (15 marks)**

Indicative mark scheme and model answer.
1. Proposed topology [4 marks]. An **extended star (hierarchical star)**: on each floor, an extended star with a floor switch (access switch) in the wiring closet, each office's 2 data points cabled individually to it; the three floor switches are linked by a backbone to a central core switch (star of stars). Wireless access points on each floor connect to the floor switch.



Extended star (hierarchical) topology for a three-floor building

Marking: 1 mark for star per floor, 1 for backbone star to a core, 1 for labelled diagram, 1 for including APs and both data points per office.

2. Central equipment [4 marks]. Use managed switches at both access (floor) and core levels — not hubs. Justification:

- Each switch port is a separate **collision domain**, eliminating collisions and giving full bandwidth per office (a hub would create one large collision domain).
- Managed switches support **VLANs**, so each department gets its own **broadcast domain**, improving security and reducing broadcast traffic.
- Manageability: SNMP monitoring, port security, remote configuration — required for centralised management.

Marking: 1 mark for choosing managed switches, 1 for collision domains, 1 for broadcast domains/VLANs, 1 for manageability.

3. Cabling [3 marks].

- Horizontal cabling (offices to floor closet): UTP Cat 6 (or Cat 6A) — supports 1 Gbps over the full 100 m and 10 Gbps up to about 55 m, cheap and easy to terminate; runs here are short (one floor).
- Backbone (floor to core): fibre optic (e.g., OM3/OM4 multimode) — supports 10 Gbps and beyond over the inter-floor distances, immune to electromagnetic interference, and satisfies the "future 10 Gbps backbone" requirement. (Cat 6A copper is acceptable only if distances stay under 100 m, but fibre is the future-proof answer.)

Marking: 1 mark per correct choice with bandwidth/distance justification (3 marks total).

4. Core redundancy [2 marks]. Deploy two core switches with each floor switch dual-homed (one uplink to each core), using link aggregation and a redundancy protocol (e.g., spanning tree with rapid convergence, or stacked/HSRP-style first-hop redundancy). If one core fails, traffic automatically flows through the second. (1 mark for dual core/dual uplinks, 1 for the failover mechanism.)

5. Regulatory/standards bodies [2 marks]. Any two of:

- ANTIC (National Agency for Information and Communication Technologies) — ICT security and standards guidelines in Cameroon.
- ART (Telecommunications Regulatory Board) — regulation of telecommunications installations.
- International standards bodies also accepted if justified: ISO/IEC and TIA/EIA (structured cabling standards such as TIA-568).

(1 mark each, any two valid bodies.)

SOLUTION **Exercise 10 — Equipment selection and cost analysis (12 marks)**

1. Available ports [2 marks].

- Proposal A: cascading uses 1 port on each switch for the inter-switch link: (24 - 1) + (8 - 1) = 23 + 7 = 30 ports for workstations.
- Proposal B: no inter-switch link needed: 48 ports available.

2. Comparison table [5 marks — 1 per correct row].

Criterion	Proposal A (unmanaged)	Proposal B (managed PoE+)
(i) Collision domains	One per switch port (switches), but no control; cascaded link is a bottleneck	One per port, with configurable settings
(ii) Broadcast domains	Single broadcast domain for all 25 PCs	Can be segmented into several via VLANs
(iii) VLAN capability	None (unmanaged)	Yes — departmental VLANs, security
(iv) QoS for VoIP	Not supported; voice and data compete equally	QoS queues prioritise voice traffic
(v) PoE power budget	None — phones/APs need separate power adapters	PoE+ powers phones and APs directly from the switch

3. Scalability [2 marks]. Proposal B scales better: 48 - 25 = 23 free ports, enough for the 15 extra workstations with 8 spare, with no new equipment. Proposal A has only 30-25 = 5 free ports, so a third switch (and another cascade hop, adding latency and a failure point) would be required. (1 mark for the port arithmetic, 1 for the conclusion.)

4. Initial cost [1 mark].

Proposal A : 85 000 + 25 000 = 110 000 FCFA

Proposal B : 420 000 FCFA

5. Two technical reasons to prefer B [2 marks — 1 each]. Any two of:

- Manageability: remote monitoring/configuration (SNMP), port security, faster fault diagnosis — essential for a public institution.
- VLAN segmentation: separates departments/services into distinct broadcast domains, improving security and performance.
- PoE+: powers VoIP phones and Wi-Fi access points without separate electrical installations.
- No cascade bottleneck/single point of failure between two switches; one device is simpler and more reliable.

SOLUTION **Exercise 11 — Practical configuration and troubleshooting (13 marks)**

1. Configuration of Router A [5 marks — 1 per logical block].

- Enter privileged and global configuration modes: `enable` then `configure terminal`
- LAN interface (e.g., GigabitEthernet0/0): `interface g0/0 ; ip address 192.168.10.1 255.255.255.0 ; no shutdown`
- Serial interface, DCE side (must provide the clock): `interface s0/0/0 ; ip address 10.0.0.1 255.255.255.252 ; clock rate 64000 ; no shutdown`
- Static route to LAN B (via Router B's serial address 10.0.0.2): `ip route 192.168.20.0 255.255.255.0 10.0.0.2`
- Save: `end` then `copy running-config startup-config`

Examiner expectations: correct subnet masks (/24 and /30), *no shutdown* on both interfaces, and *clock rate* only on the DCE side are the classic marking points.

2. "Protocol down" on the serial link — three Layer 1/2 causes [3 marks — 1 each].

- Missing clock rate on the DCE side — verify with `show controllers serial0/0/0` (shows DCE/DTE and clocking).
- Encapsulation mismatch (HDLC on one side, PPP on the other) — verify with `show interfaces serial0/0/0` (line "Encapsulation").
- Interface shut down or faulty/loose serial cable — verify with `show ip interface brief` (status "administratively down") and by physically inspecting/replacing the cable.

3. Ping works but HTTP (port 80) fails — two Layer 4+ causes [3 marks].

- The web server service is not running on PC2 (or listening on a different port). Test: on PC2 run `netstat -an` and check for a listener on port 80, or from PC1 try `telnet 192.168.20.x 80` — connection refused confirms the service is down.
- An access control list (ACL) on a router blocks TCP port 80 while permitting ICMP.

Test: `show access-lists` and `show ip interface` on both routers to see applied ACLs; remove or correct the offending entry.

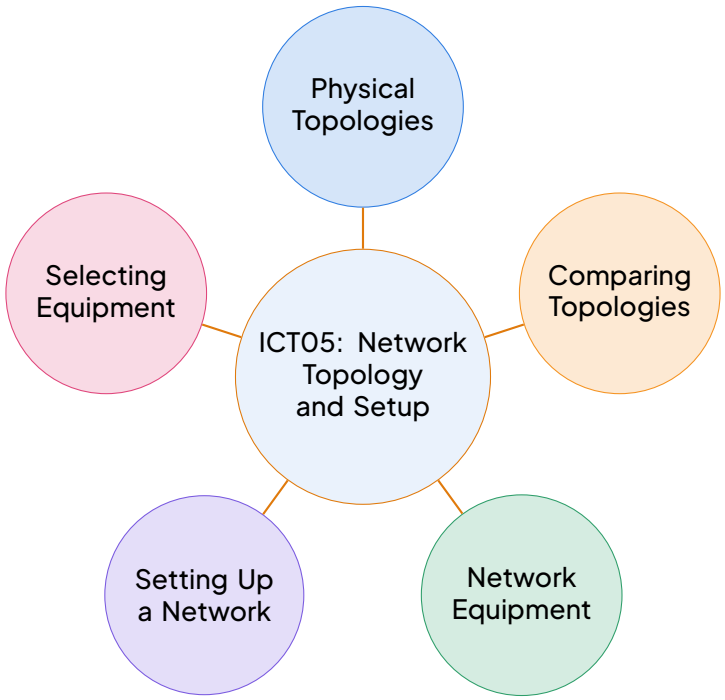
(1 mark per cause, 1 mark for valid tests; other valid answers, e.g., proxy misconfiguration, accepted.)

4. T568B pinout for the straight-through patch cables [2 marks].

Pin 1: White/Orange — Pin 2: Orange — Pin 3: White/Green — Pin 4: Blue — Pin 5: White/Blue — Pin 6: Green — Pin 7: White/Brown — Pin 8: Brown, identical on both ends.

(2 marks for the complete correct order; 1 mark if one or two pairs are swapped.)

20 SUMMARY SHEET



Mind map of Chapter ICT05

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key Definitions

- **Physical Topology**: The actual layout of cables and devices in a network.
- **Logical Topology**: The way data flows between devices, independent of physical layout.
- **Network Equipment**: Hardware devices (switch, router, access point, modem, firewall) that enable communication.
- **Structured Cabling**: Standardised cabling system (TIA/EIA-568) for reliable connectivity.

Main Topologies & Characteristics

- Bus: Single backbone cable, easy to install, but a break disables whole network.
- Star: Central switch/hub, fault isolation, scalable, most common in LANs.
- Ring: Each node connects to two neighbours, token passing, deterministic but single point of failure.

- Mesh: Every node connects to many others, high redundancy, costly, used in backbone/WAN.
- Hybrid: Combination (e.g., star-ring, star-bus) to balance cost and reliability.

Choosing a Topology – Decision Criteria

- Cost (cabling, ports, equipment)
- Scalability (adding nodes)
- Fault tolerance (single point of failure)
- Performance (bandwidth, latency)
- Management complexity
- Environment (office, campus, industrial)

Key Network Equipment

- **Switch** (Layer 2): Forwards frames based on MAC address; creates collision domains per port.
- **Router** (Layer 3): Connects different networks, routes packets by IP address.
- **Access Point**: Provides wireless connectivity (Wi-Fi 5/6/6E).
- **Modem**: Converts digital signals for ISP link (DSL, fibre, 4G/5G).
- **Firewall**: Filters traffic based on security policies.
- **Patch Panel / Keystone**: Terminates structured cabling.

Setting Up a LAN – Step-by-Step Method

1. Plan: Draw topology, list devices, estimate cable lengths.
2. Procure: Switch(es), router, cables (Cat6/6a), patch panels, rack, UPS.
3. Install structured cabling: Run horizontal cables to outlets, terminate on patch panel.
4. Connect devices: Patch panel → switch ports; switch uplink → router WAN.
5. Configure: Assign static IPs or DHCP scope, VLANs, port security, SSID/WPA3 on AP.
6. Test: Ping, traceroute, cable certifier, throughput test (iperf3).
7. Document: Label cables, update network diagram, record IP scheme.

Selecting Equipment for an Organisation

- Small office (≤20 users): Unmanaged switch, SOHO router, single AP, Cat6.
- Medium enterprise (20–200): Managed switches (VLAN, QoS, PoE+), dual-WAN router, controller-based APs, core switch with stacking.

- Large campus (>200): Core/distribution/access layer, Layer 3 switches, redundant links (STP/RSTP, VRRP), wireless controller cluster, firewall with IPS/IDS, network management (SNMP, NetFlow).

Common Mistakes (GCE Traps)

- Confusing physical vs. logical topology.
- Thinking a hub is a switch (hub = shared collision domain).
- Forgetting that a star topology still has a single point of failure at the central switch.
- Misidentifying cable categories (Cat5e vs Cat6 vs Cat6a) and max lengths (100 m).
- Neglecting to configure VLANs on a managed switch before connecting devices.
- Assuming wireless is always slower; modern Wi-6 can exceed 1 Gbps.

GCE Examination Tips

- Know the diagram symbols for each topology and be able to draw them.
- Be ready to compare two topologies in a table (cost, fault tolerance, scalability).
- Questions often ask: "Which topology for a school lab with 30 PCs?" → Star with managed switch.
- Understand the OSI layers of each device: Switch (L2), Router (L3), Firewall (L3/L4), AP (L1/L2).
- Practise writing a short network setup procedure (5–7 steps).
- Remember Cameroon context: MTN/Orange fibre, CAMTEL backbone, ANTIC regulations for cybersecurity.

Signed • OnBuch+
ONBUCH+ TEACHING TEAM
FOLLOWS THE OFFICIAL MINESEC SYLLABUS

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